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CREATE CHANGE

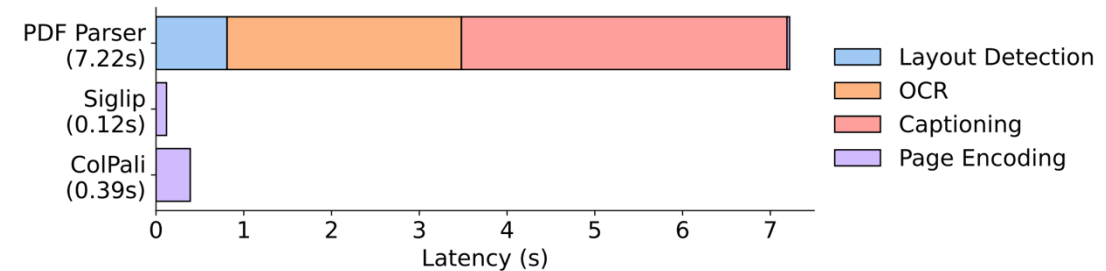
Multimodal RAG Evaluation and Hallucinations

INFS 4205/7205 Advanced Techniques for High Dimensional Data

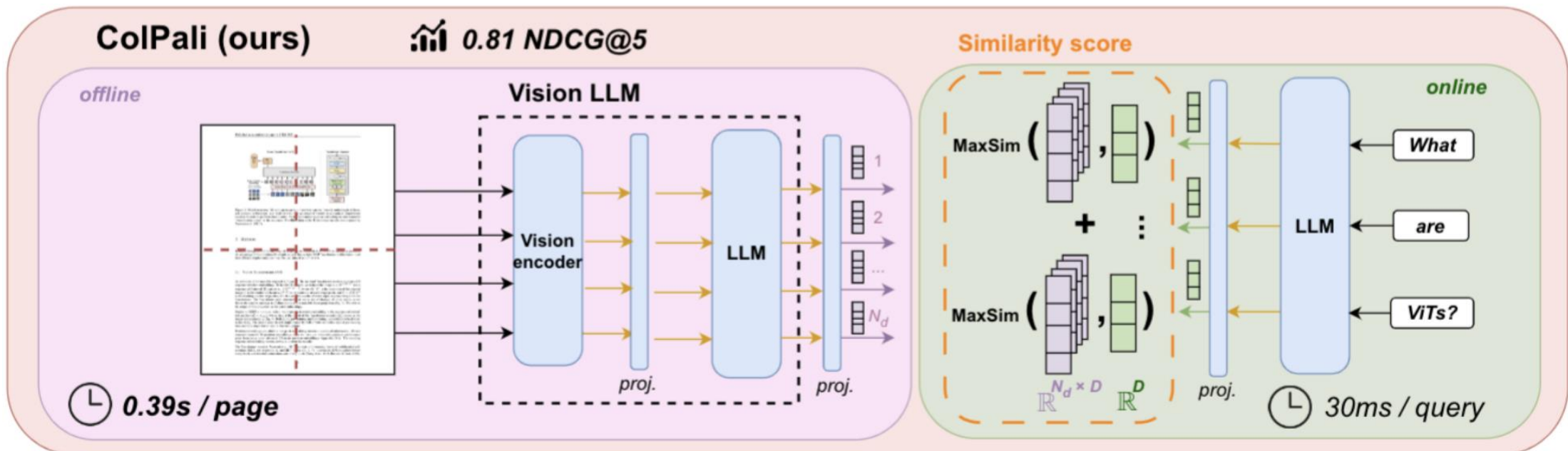
Yadan Luo

School of EECS, The University of Queensland

- multimodal document retrieval is not the same as OCR-only search
- **Key idea:** page images can be embedded directly and matched with *late interaction*



Today's question: if retrieval is correct, is generated answer also correct?

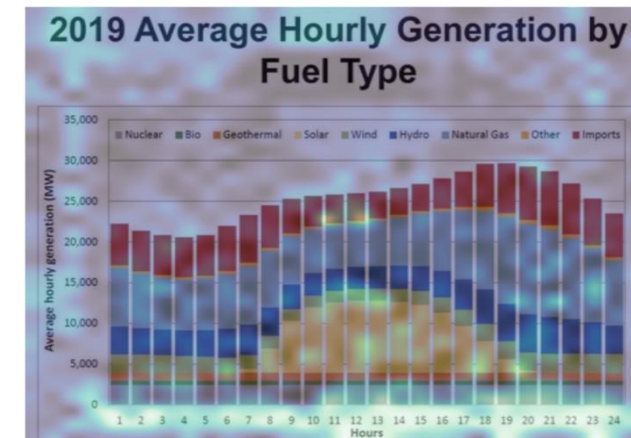


Recap: ColPali

Retrieval is necessary, not sufficient...

	ArxivQ	DocQ	InfoQ	TabF	TATQ	Shift	AI	Energy	Gov.	Health.	Avg.
Unstructured text-only											
- BM25	-	34.1	-	-	44.0	59.6	90.4	78.3	78.8	82.6	-
- BGE-M3	-	28.4 $\downarrow$ 5.7	-	-	36.1 $\downarrow$ 7.9	68.5 $\uparrow$ 8.9	88.4 $\downarrow$ 2.0	76.8 $\downarrow$ 1.5	77.7 $\downarrow$ 1.1	84.6 $\uparrow$ 2.0	-
Unstructured + OCR											
- BM25	31.6	36.8	62.9	46.5	62.7	64.3	92.8	85.9	83.9	87.2	65.5
- BGE-M3	31.4 $\downarrow$ 0.2	25.7 $\downarrow$ 11.1	60.1 $\downarrow$ 2.8	70.8 $\uparrow$ 24.3	50.5 $\downarrow$ 12.2	73.2 $\uparrow$ 8.9	90.2 $\downarrow$ 2.6	83.6 $\downarrow$ 2.3	84.9 $\uparrow$ 1.0	91.1 $\uparrow$ 3.9	66.1 $\uparrow$ 0.6
Unstructured + Captioning											
- BM25	40.1	38.4	70.0	35.4	61.5	60.9	88.0	84.7	82.7	89.2	65.1
- BGE-M3	35.7 $\downarrow$ 4.4	32.9 $\downarrow$ 5.4	71.9 $\uparrow$ 1.9	69.1 $\uparrow$ 33.7	43.8 $\downarrow$ 17.7	73.1 $\uparrow$ 12.2	88.8 $\uparrow$ 0.8	83.3 $\downarrow$ 1.4	80.4 $\downarrow$ 2.3	91.3 $\uparrow$ 2.1	67.0 $\uparrow$ 1.9
Contrastive VLMs											
Jina-CLIP	25.4	11.9	35.5	20.2	3.3	3.8	15.2	19.7	21.4	20.8	17.7
Nomic-vision	17.1	10.7	30.1	16.3	2.7	1.1	12.9	10.9	11.4	15.7	12.9
SigLIP (Vanilla)	43.2	30.3	64.1	58.1	26.2	18.7	62.5	65.7	66.1	79.1	51.4
Ours											
SigLIP (Vanilla)	43.2	30.3	64.1	58.1	26.2	18.7	62.5	65.7	66.1	79.1	51.4
BiSigLIP (+fine-tuning)	58.5 $\uparrow$ 15.3	32.9 $\uparrow$ 2.6	70.5 $\uparrow$ 6.4	62.7 $\uparrow$ 4.6	30.5 $\uparrow$ 4.3	26.5 $\uparrow$ 7.8	74.3 $\uparrow$ 11.8	73.7 $\uparrow$ 8.0	74.2 $\uparrow$ 8.1	82.3 $\uparrow$ 3.2	58.6 $\uparrow$ 7.2
BiPali (+LLM)	56.5 $\downarrow$ 2.0	30.0 $\downarrow$ 2.9	67.4 $\downarrow$ 3.1	76.9 $\uparrow$ 14.2	33.4 $\uparrow$ 2.9	43.7 $\uparrow$ 17.2	71.2 $\downarrow$ 3.1	61.9 $\downarrow$ 11.7	73.8 $\downarrow$ 0.4	73.6 $\downarrow$ 8.8	58.8 $\uparrow$ 0.2
ColPali (+Late Inter.)	79.1 $\uparrow$ 22.6	54.4 $\uparrow$ 24.5	81.8 $\uparrow$ 14.4	83.9 $\uparrow$ 7.0	65.8 $\uparrow$ 32.4	73.2 $\uparrow$ 29.5	96.2 $\uparrow$ 25.0	91.0 $\uparrow$ 29.1	92.7 $\uparrow$ 18.9	94.4 $\uparrow$ 20.8	81.3 $\uparrow$ 22.5

- Retrieval results are presented using **nDCG@5** metrics



Query: "Which hour of the day had the highest overall electricity generation in 2019?"



ViDoRe: Visual Document Retrieval

ViDoRe V3: our new benchmark release!

Developed by ILLUIN Technology and with contributions from NVIDIA, it is the most diverse visual document retrieval benchmark to date for enterprise applications. It includes 10 datasets, 26,000+ pages, 3000+ queries in 6 languages. Check out: [the paper](#) or [the blogpost](#)!

This organization contains models, datasets, benchmarks and code released with the ViDoRe project by Illuin Technology.

• **Leaderboard:**

• [ViDoRe Leaderboard](#)

• **Benchmarks:**

• ViDoRe V1 ([paper](#), [blogpost](#), [dataset collection](#))

• ViDoRe V2 ([paper](#), [blogpost](#), [dataset collection](#))

• ViDoRe V3 ([paper](#), [blogpost](#), [dataset collection](#))

• **Models:**

Recap: Generation Difficulties

Retrieval is necessary, not sufficient...

- Correct evidence can still lead to an *incorrect* answer
- The generator may *ignore* evidence, *over-trust* priors, or *fuse* modalities badly

2. Generation Difficulties

Retriever

Generator (LLM)

M-RAG therefore needs layered evaluation!
(not only retrieval or final-answer evaluation)

"How much is the fee
for a H-1B visa
in 2026?"

User Question

"The fee for H-1B
in 2026 is \$15,000."

Hallucinated Response

Generation Difficulties

[Knowledge base]

• Relevant document ✓

By 2026, a new one-time \$100,000 payment is imposed on all new H-1B petitions, in addition to the existing fees ...

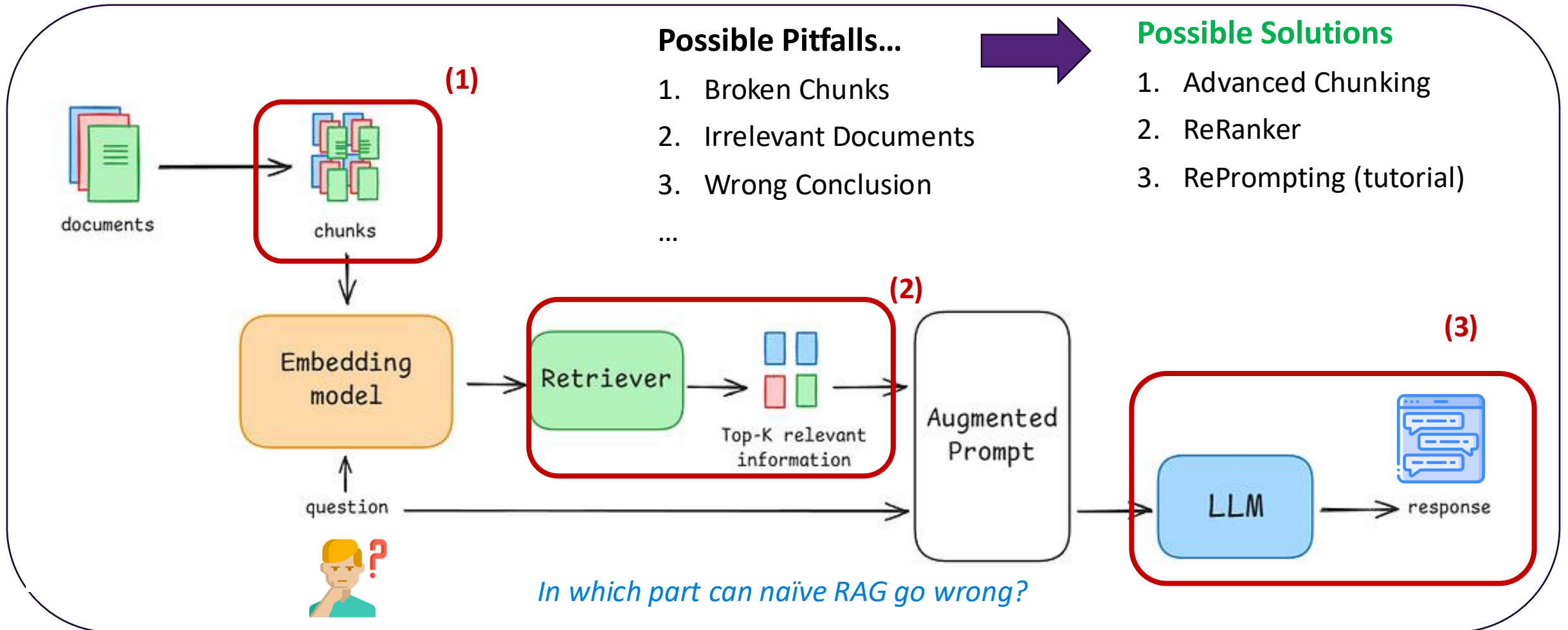
• Relevant document ✓

In 2024, the H-1B visa process requires a \$205 consular application fee along with multiple USCIS petition fees ...

Irrelevant document

By 2026, U.S. technology, biotech, and semiconductor sectors are projected to depend heavily on H-1B professionals...

Recap: Multimodal RAG



RAG is a multi-stage pipeline, meaning **failure** can happen at the retrieval or generation stage.

RAGAs: Retrieval Augmented Generation Assessment

Ragas Office Hours - If you need help setting up Evals for your AI application, sign up for our Office Hours [here](#).

Ragas Search

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Get Started

Installation

Quick Start

Tutorials

Evaluate a prompt

Evaluate a simple RAG system

Evaluate an AI Workflow

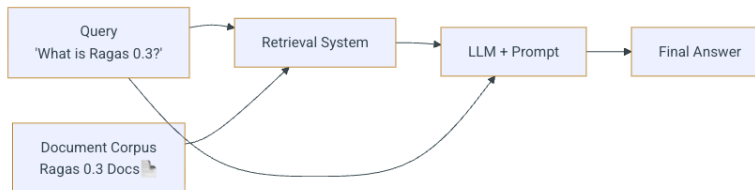
Evaluate an AI Agent

Get Started > Tutorials

Evaluate a simple RAG system

Copy page

In this tutorial, we will write a simple evaluation pipeline to evaluate a RAG (Retrieval-Augmented Generation) system. At the end of this tutorial, you'll learn how to evaluate and iterate on a RAG system using evaluation-driven development.



We will start by writing a simple RAG system that retrieves relevant documents from a corpus and generates an answer using an LLM.

```
python -m ragas_examples.rag_eval.rag
```

Next, we will write down a few sample queries and expected outputs for our RAG system. Then convert them to a CSV file.

```
import pandas as pd

samples = [
    {"query": "What is Ragas 0.3?", "grading_notes": "- Ragas 0.3 is a library for"},
    {"query": "How to install Ragas?", "grading_notes": "- install from source -"},
    {"query": "What are the main features of Ragas?", "grading_notes": "organised"}
]

pd.DataFrame(samples).to_csv("datasets/test_dataset.csv", index=False)
```

ragas Public

Watch 47 Fork 1.4k Star 13.7k

main 3 Branches 90 Tags

Go to file Add file Code

veeceey fix: allow fork contributors in check-docs CI workflow (#2606) 298b682 · 2 months ago 1,147 Commits

.claude/commands	fix: metric inheritance patterns: separate factory-created ...	7 months ago
.cursor	docs: fix modifying-prompts-metrics guide with correct A...	4 months ago

About

Supercharge Your LLM Application Evaluations

[docs.ragas.io](#)

evaluation llm llmops

```
from openai import AsyncOpenAI
from ragas.llms import llm_factory
from ragas.metrics.collections import AnswerAccuracy

# Setup LLM
client = AsyncOpenAI()
llm = llm_factory("gpt-4o-mini", client=client)

# Create metric
scorer = AnswerAccuracy(llm=llm)

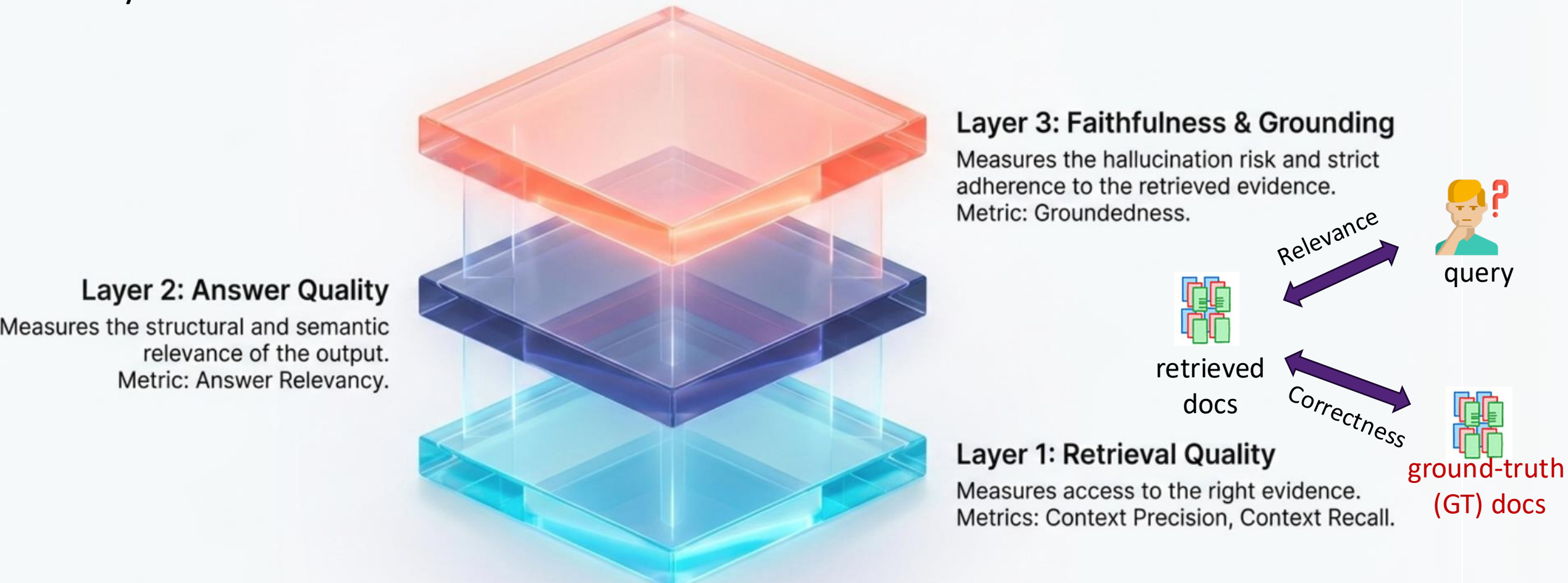
# Evaluate
result = await scorer.ascore(
    user_input="When was Einstein born?",
    response="Albert Einstein was born in 1879.",
    reference="Albert Einstein was born in 1879."
)
print(f"Answer Accuracy Score: {result.value}")
```

Output:

```
Answer Accuracy Score: 1.0
```

RAGAs: Retrieval Augmented Generation Assessment

3-layered architecture of MRAG evaluation



RAGAs: Retrieval Metrics

Retrieval metrics are still the first layer of evaluation in M-RAG

- *Non-Rank Based Metrics*: often assess **binary** outcomes

- Accuracy?

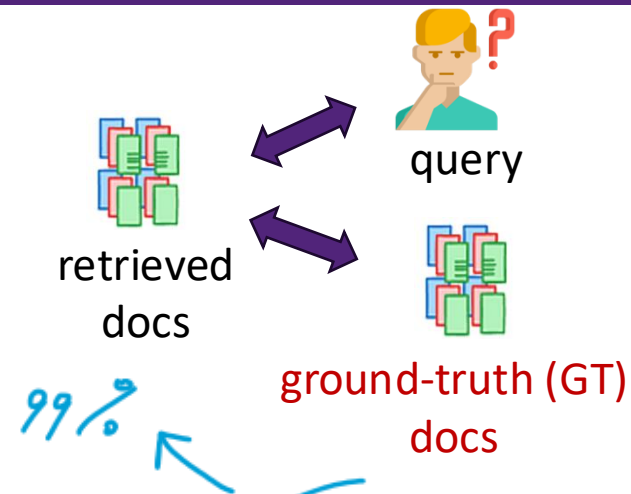
$$\text{accuracy} = (tp + tn) / (tp + fp + fn + tn)$$

- Precision, Recall

fraction of retrieved documents that are relevant

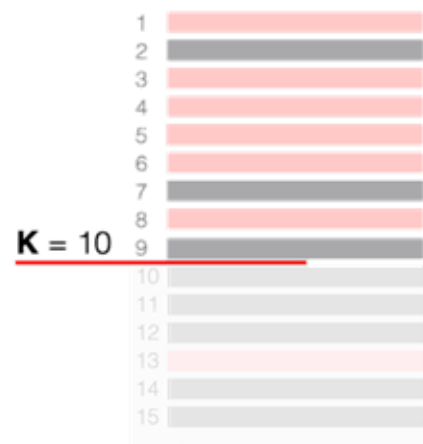
$$\text{Precision} = \frac{\#(\text{relevant items retrieved})}{\#(\text{retrieved items})} = P(\text{relevant}|\text{retrieved})$$

$$P = tp / (tp + fp)$$

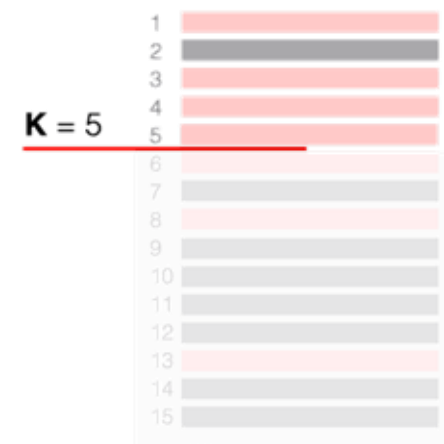


	Relevant	Nonrelevant
Retrieved	true positives (tp)	false positives (fp)
Not retrieved	false negatives (fn)	true negatives (tn)

Precision@10 = 0.6



Precision@5 = 0.8



RAGAs: Retrieval Metrics

Retrieval metrics are still the first layer of evaluation in M-RAG

- **Non-Rank Based Metrics:** often assess **binary** outcomes

- **Accuracy?**

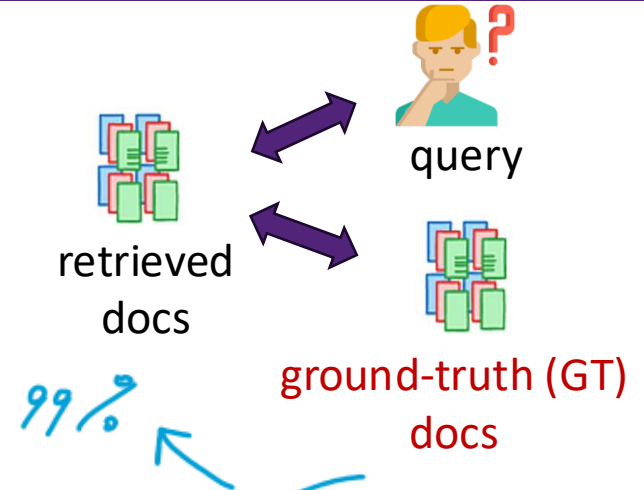
$$\text{accuracy} = (tp + tn) / (tp + fp + fn + tn)$$

- **Precision, Recall**

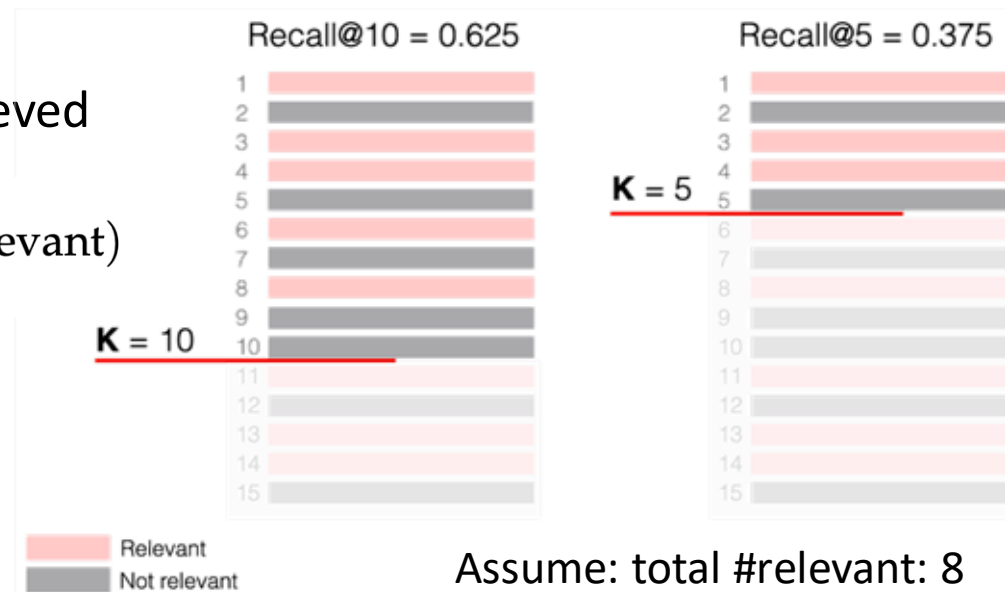
fraction of relevant documents that are retrieved

$$\text{Recall} = \frac{\#(\text{relevant items retrieved})}{\#(\text{relevant items})} = P(\text{retrieved}|\text{relevant})$$

$$R = tp / (tp + fn)$$



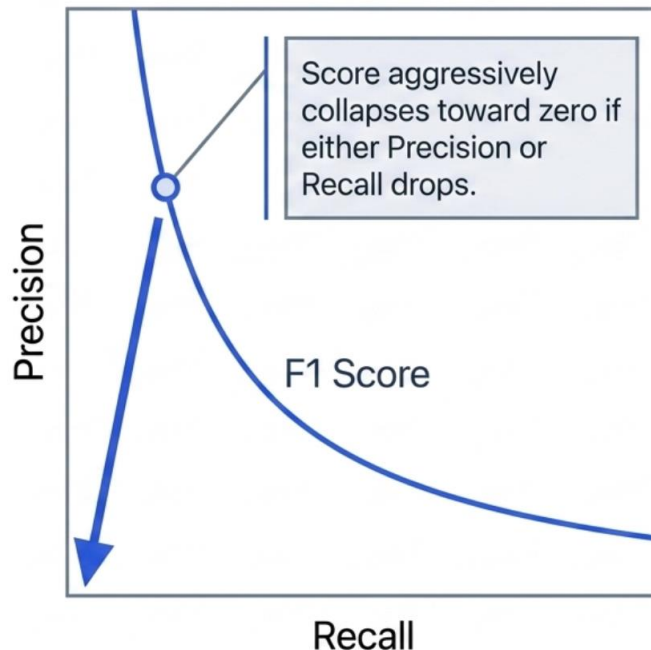
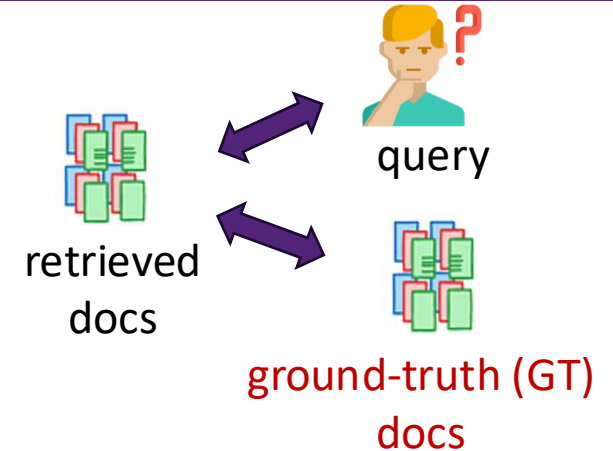
	Relevant	Nonrelevant
Retrieved	true positives (tp)	false positives (fp)
Not retrieved	false negatives (fn)	true negatives (tn)



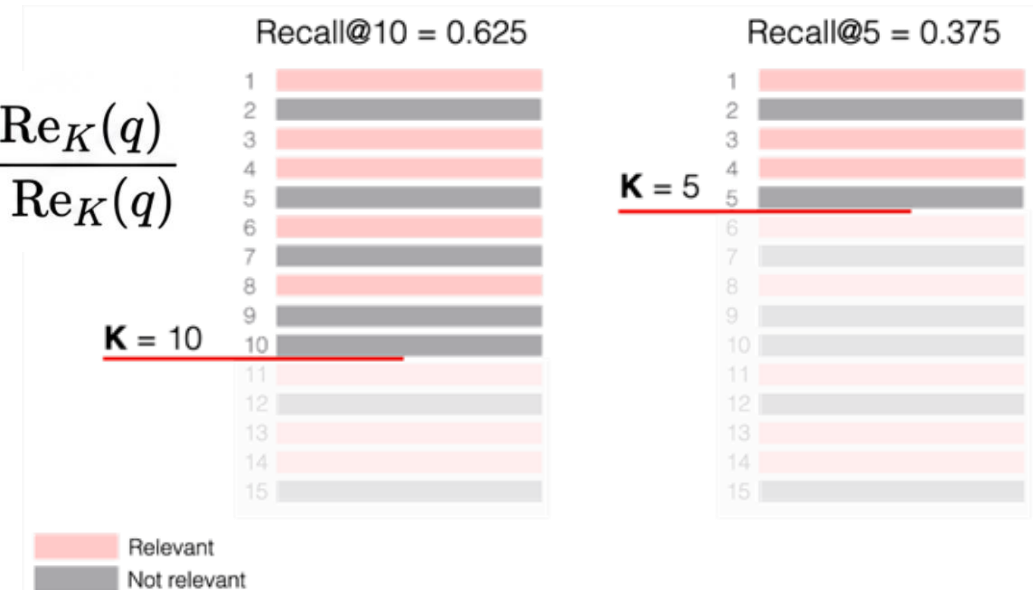
RAGAs: Retrieval Metrics

Retrieval metrics are still the first layer of evaluation in M-RAG

- *Non-Rank Based Metrics*: often assess **binary** outcomes
- **F1: Harmonic mean of P and R**



$$F1@K = 2 \cdot \frac{\text{Pr}_K(q) \cdot \text{Re}_K(q)}{\text{Pr}_K(q) + \text{Re}_K(q)}$$

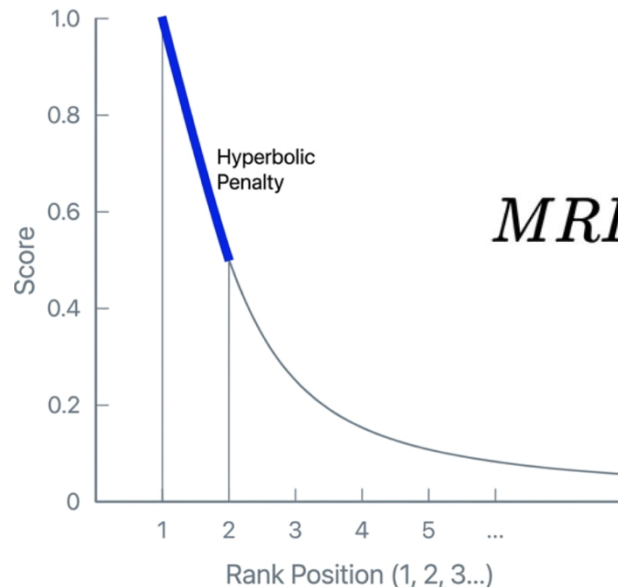
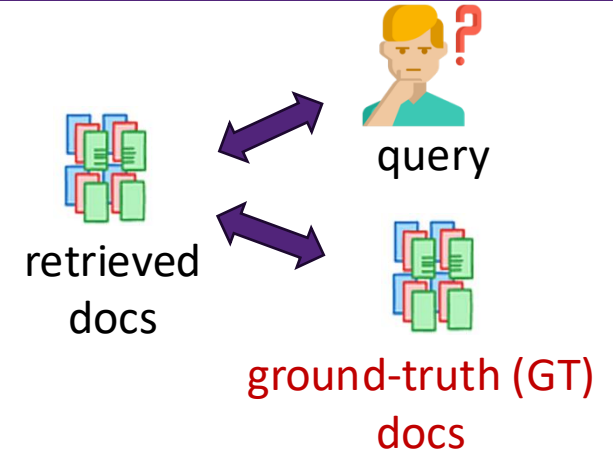


Retrieval model can “cheat” recall by increasing K...

RAGAs: Retrieval Metrics

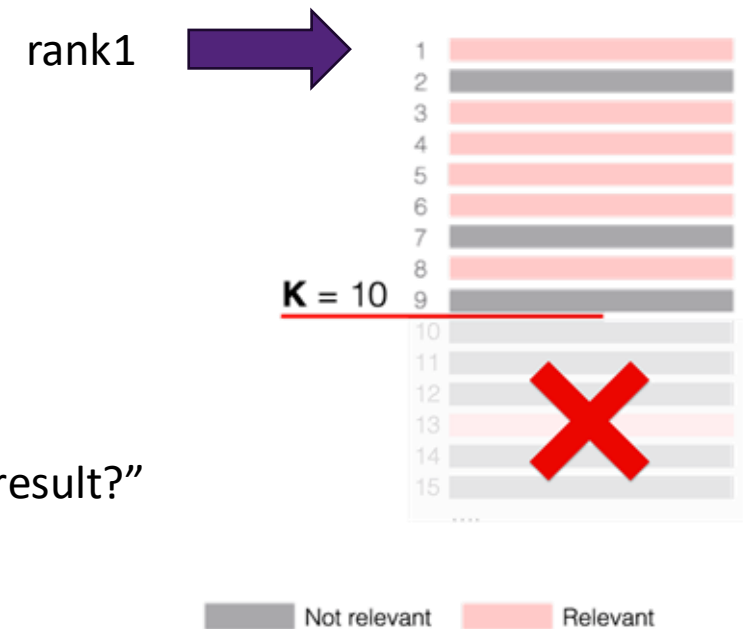
Retrieval metrics are still the first layer of evaluation in M-RAG

- **Rank Based Metrics:** evaluate the order in which relevant items are shown
- **Mean Reciprocal Rank (MRR)**
 - average of the reciprocal ranks of the first correct answer for a set of queries.



$$MRR = \frac{1}{Q} \sum_{q=1}^Q \frac{1}{rank_q}$$

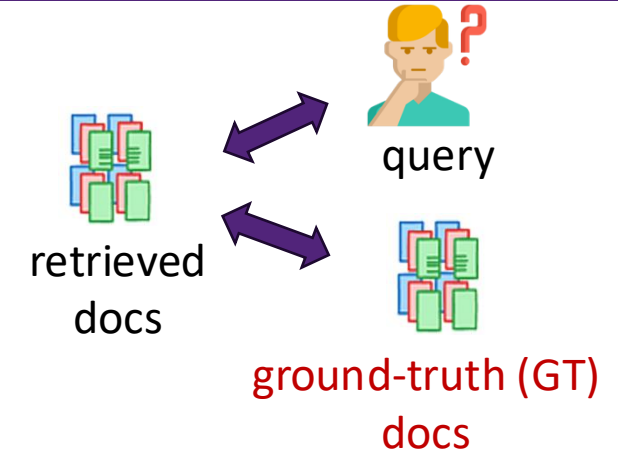
=> "How early is the first relevant result?"



RAGAs: Retrieval Metrics

Retrieval metrics are still the first layer of evaluation in M-RAG

- **Rank Based Metrics:** evaluate the order in which relevant items are shown
- **Normalised discounted cumulative gain (nDCG@K)**
 - designed for situations of non-binary notions of relevance -> **graded relevance**



$$\text{NDCG@K} = \frac{\text{DCG@K}}{\text{IDCG@K}}$$

Ideal ranking



vs.

Real ranking



=> "Are the most relevant results ranked highest?"

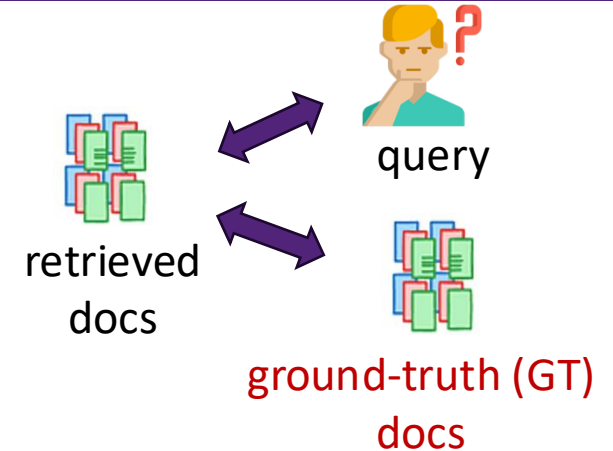
Not relevant Very relevant

RAGAs: Retrieval Metrics

Normalised discounted cumulative gain (nDCG@K)

- designed for situations of **non-binary** notions of relevance

$$\text{NDCG@K} = \frac{\text{DCG@K}}{\text{IDCG@K}}$$



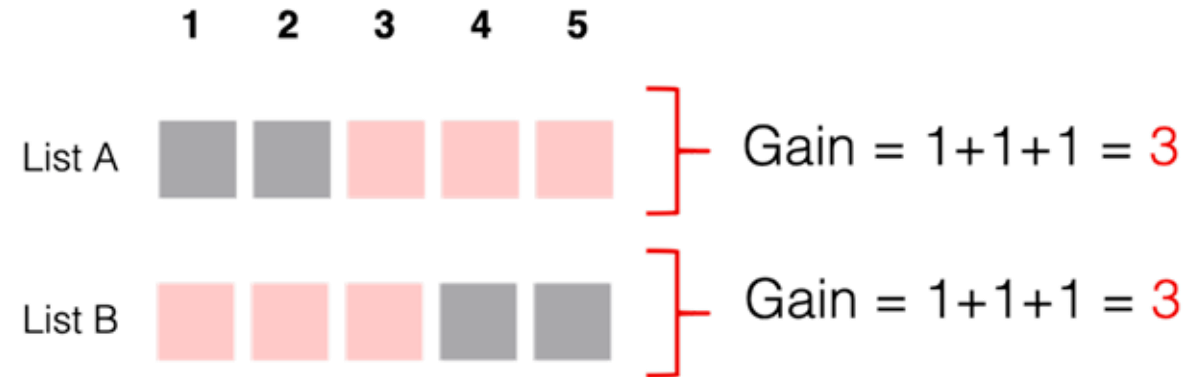
Cumulative Gain

- measure of the total relevance of the ranked list.
 - > sum(relevance scores)
- Can be *binary* or *graded*

Discounted Cumulative Gain (DCG)

$$\text{DCG} = \sum_{i=1}^n \frac{rel_i}{\log_2(i+1)}$$

A *logarithmic* discount helps assign *lower* gain when relevant items appear further down in the ranked list.

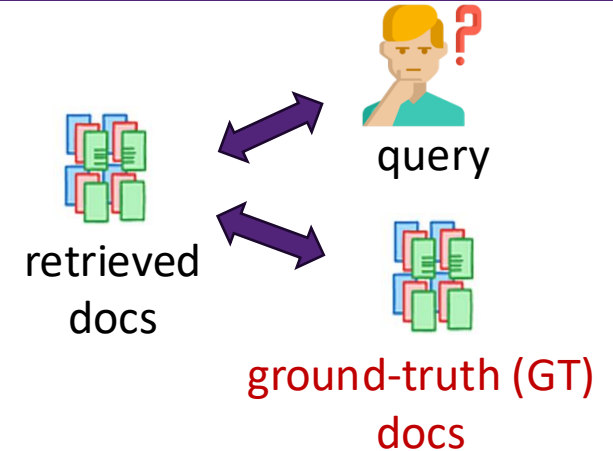


RAGAs: Retrieval Metrics

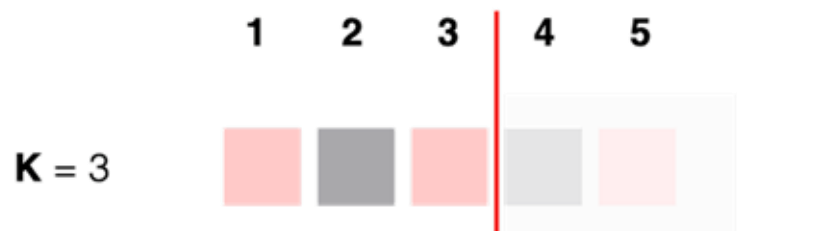
Normalised discounted cumulative gain (nDCG@K)

- designed for situations of **non-binary** notions of relevance

$$\text{NDCG@K} = \frac{\text{DCG@K}}{\text{IDCG@K}}$$



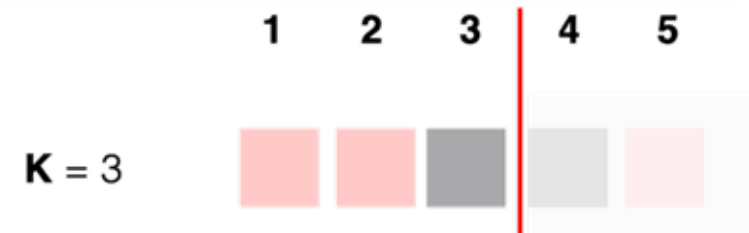
Discounted Cumulative Gain (DCG)



$$\text{DCG@3} = \frac{1}{\log_2(1+1)} + \frac{0}{\log_2(1+2)} + \frac{1}{\log_2(1+3)}$$

$$\text{DCG@3} = 1 + 0 + 0.5 = 1.5$$

$$\text{DCG} = \sum_{i=1}^n \frac{rel_i}{\log_2(i+1)}$$



$$\text{DCG@3} = \frac{1}{\log_2(1+1)} + \frac{1}{\log_2(1+2)} + \frac{0}{\log_2(1+3)}$$

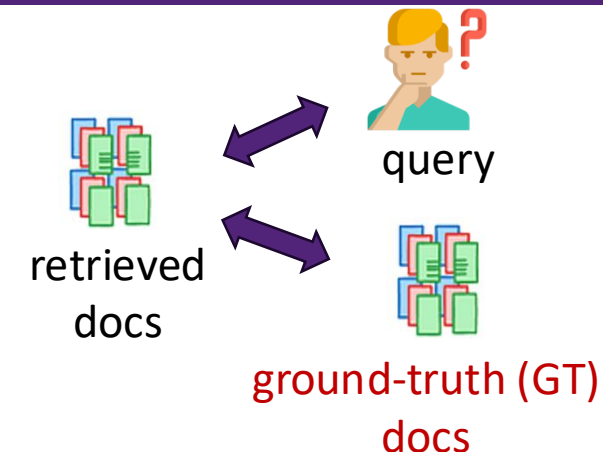
$$\text{DCG@3} = 1 + \frac{1}{1.585} + 0 \approx 1.63$$

RAGAs: Retrieval Metrics

Normalised discounted cumulative gain (nDCG@K)

- designed for situations of **non-binary** notions of relevance

$$\text{NDCG@K} = \frac{\text{DCG@K}}{\text{IDCG@K}}$$



Ideal Discounted Cumulative Gain (IDCG)

$$\text{IDCG@K} = \sum_{i=1}^K \frac{rel_i^*}{\log_2(i+1)}$$



$$\text{DCG@3} = \frac{2}{\log_2(1+1)} + \frac{0}{\log_2(2+1)} + \frac{1}{\log_2(3+1)} = 2.5$$



$$\text{IDCG@3} = \frac{2}{\log_2(1+1)} + \frac{1}{\log_2(2+1)} + \frac{0}{\log_2(3+1)} \approx 2.63$$

$$\text{NDCG@3} = \frac{\text{DCG@3}}{\text{IDCG@3}} = \frac{2.5}{2.63} \approx 0.95$$

RAGAs: Retrieval Metrics

When ground-truth documents are not available:
retrieved documents -> input query (Context Relevance)

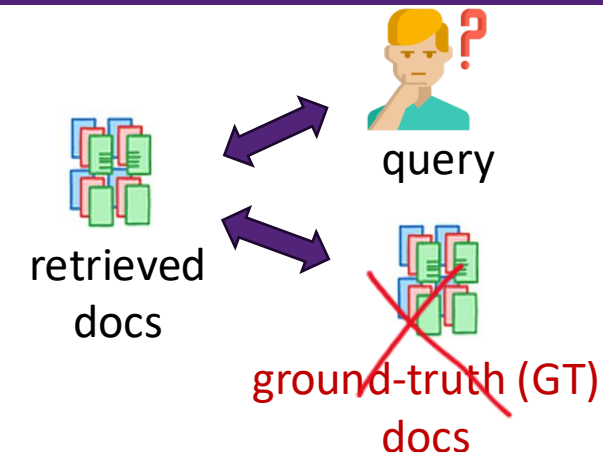
Human Annotation

- Is the document relevant to the input query?
- While it's taxing, it potentially aligns with human judgments.

Example query

How do I reset my password?

Chunk Content (Shortened)	Relevance
"To reset your password, go to Account Settings and click 'Forgot Password'..."	✓ Relevant
"If you're locked out of your account, you can recover access by verifying your email."	⚠ Partially Relevant
"To change your payment method, navigate to Billing and Payments."	✗ Not Relevant
"Password requirements: minimum 8 characters, at least one number..."	✓ Relevant
"For billing disputes, please contact support directly."	✗ Not Relevant



Automated evaluation with LLM (LLM as a Judge)

- Direct Prompting
- Prompting + ICL (e.g., format the examples on the left side into ICL)

Is the context relevant to the query?

context: {retrieved document}
 query: {query}

Is the context relevant to the query?

Example #1:
 context: To reset your password, please go to
 query: How do I reset my password?
 answer: Relevant

Example #2:

Example #3:

context: {retrieved document}
 query: {query}
 answer:

RAGAs: Retrieval Metrics

When ground-truth documents are not available:
retrieved documents -> input query (Context Relevance)

Example

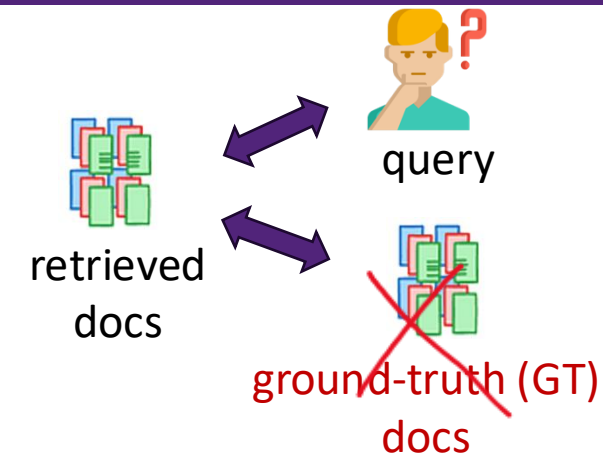
```
from openai import AsyncOpenAI
from ragas.llms import llm_factory
from ragas.metrics.collections import ContextRelevance

# Setup LLM
client = AsyncOpenAI()
llm = llm_factory("gpt-4o-mini", client=client)

# Create metric
scorer = ContextRelevance(llm=llm)

# Evaluate
result = await scorer.ascore(
    user_input="When and Where Albert Einstein was born?",
    retrieved_contexts=[
        "Albert Einstein was born March 14, 1879.",
        "Albert Einstein was born at Ulm, in Württemberg, Germany.",
    ]
)
print(f"Context Relevance Score: {result.value}")
```

In this example, the two retrieved contexts together fully address the user's query by providing both the birthdate and location of Albert Einstein. Consequently, both prompts would rate the combined contexts as **2** (fully relevant). Normalizing each score yields **1.0** (2/2), and averaging the two results maintains the final Context Relevance score at **1**.



Reranker: BGE-reranker / Cohere rerank / cross-encoder
Evaluator: RAGAS / TruLens / LLM-as-judge / human annotation

Query

Answer Relevance
Is the answer relevant to the query?

Response

Context Relevance
Is the retrieved image relevant (sufficiently similar) to the query image?

Context

Groundedness
Is the response supported by the context?

=Faithfulness

BGE-Reranker, Cohere Rerank, RankGPT...

😊 Actually, you have learned this part last week...

<https://github.com/truera/trulens>

How to measure Groundedness / Faithfulness?

- Whether the **claims** can be supported by retrieved context

Step 1: Statement Extraction (Generation Decomposition)

To evaluate Faithfulness without a human-annotated ground-truth answer, we use an LLM-as-a-Judge in a two-step pipeline. The first step is **Generation Decomposition**: breaking the final complex answer down into atomic, independent claims.

Let q be the user's query.

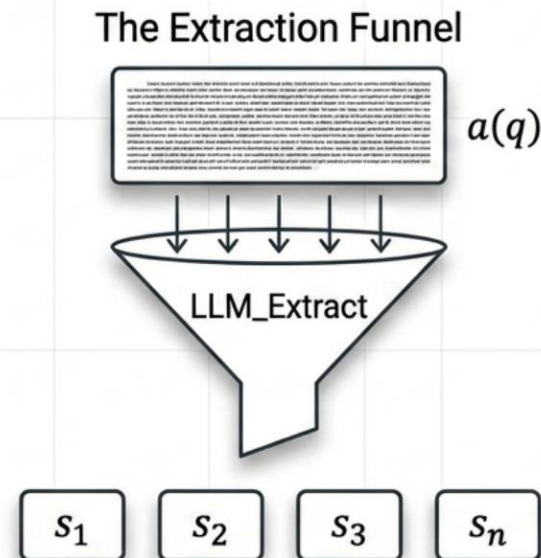
Let $c(q)$ be the retrieved context.

Let $a(q)$ be the generated answer.

Process: We prompt an LLM to parse the generated answer $a(q)$ and extract a set of independent factual statements, denoted as S .

$$S = \text{LLM_Extract}(a(q)) = \{s_1, s_2, \dots, s_n\}$$

Where n is the total number of atomic statements extracted.



How to measure Groundedness / Faithfulness?

- Whether the **claims** can be supported by retrieved context

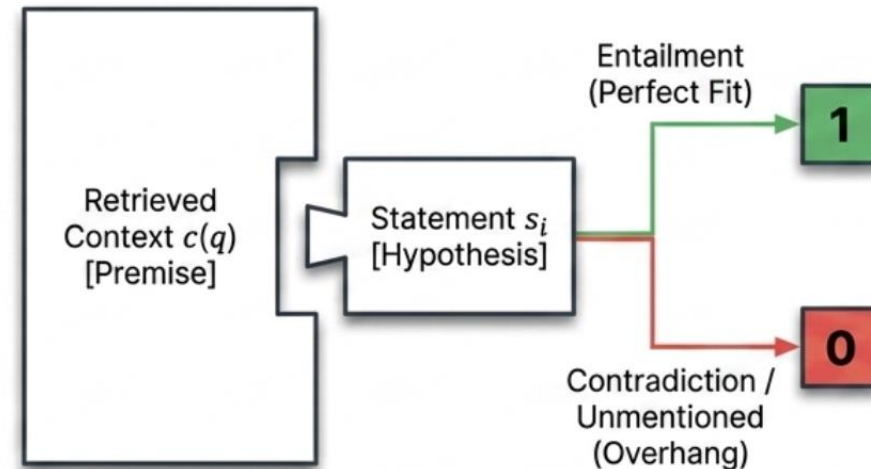
Step 2: Natural Language Inference (NLI) Verification

We treat the retrieved context as the Premise and each extracted claim as a Hypothesis. For every single statement s_i in S , we prompt an LLM Judge to determine if s_i is logically entailed by the retrieved context $c(q)$.

We define an indicator function $V(s_i, c(q))$:

$$\begin{aligned} V(s_i, c(q)) &= 1, \text{ if context } c(q) \text{ entails statement } s_i \\ V(s_i, c(q)) &= 0, \text{ otherwise (contradiction or unmentioned)} \end{aligned}$$

The Entailment Scales



How to measure Groundedness / Faithfulness?

- Whether the **claims** can be supported by retrieved context

Step 3: The Faithfulness Equation

The final Faithfulness score is the ratio of statements that are strictly supported by the context over the total number of statements generated.

$$\text{Faithfulness} = \frac{\sum V(s_i, c(q))}{|S|}$$

Total Confirmed Truths

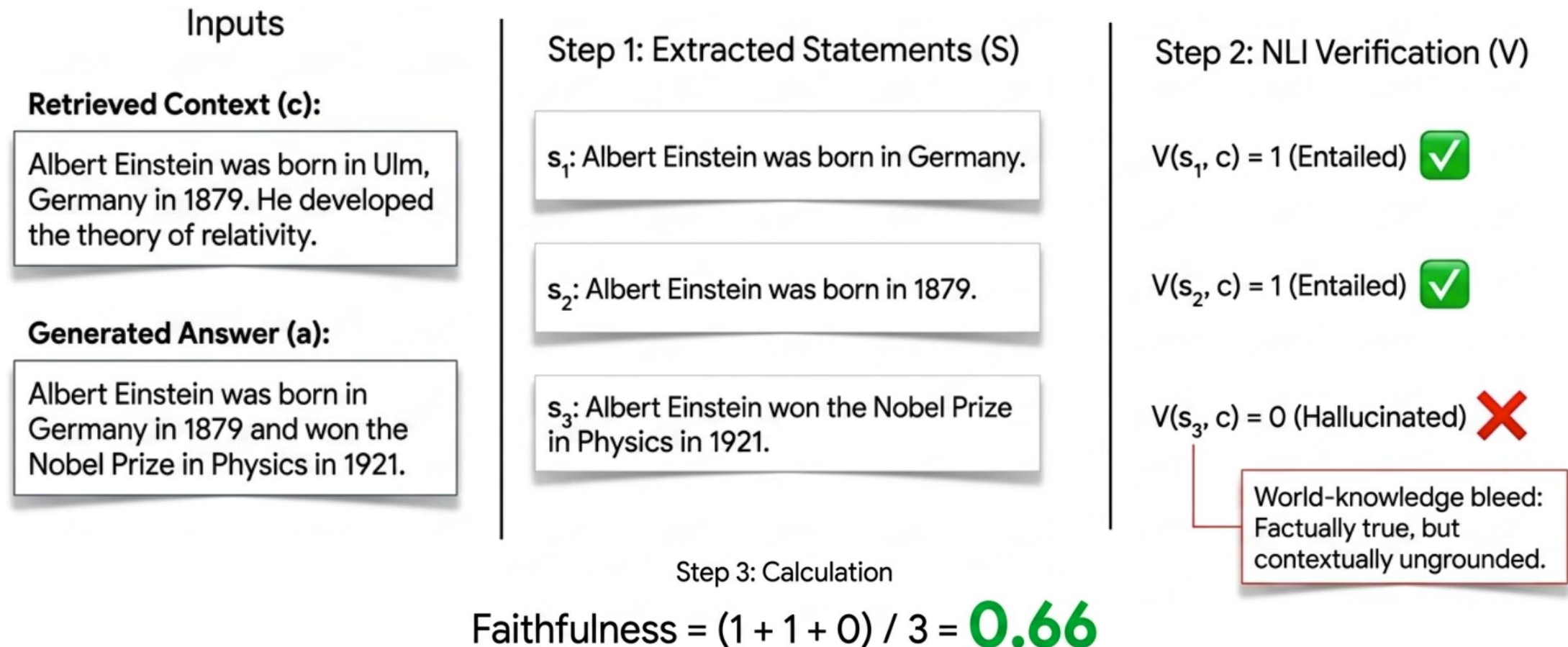
Total Claims Generated

Metric Range: [0, 1]



How to measure Groundedness / Faithfulness?

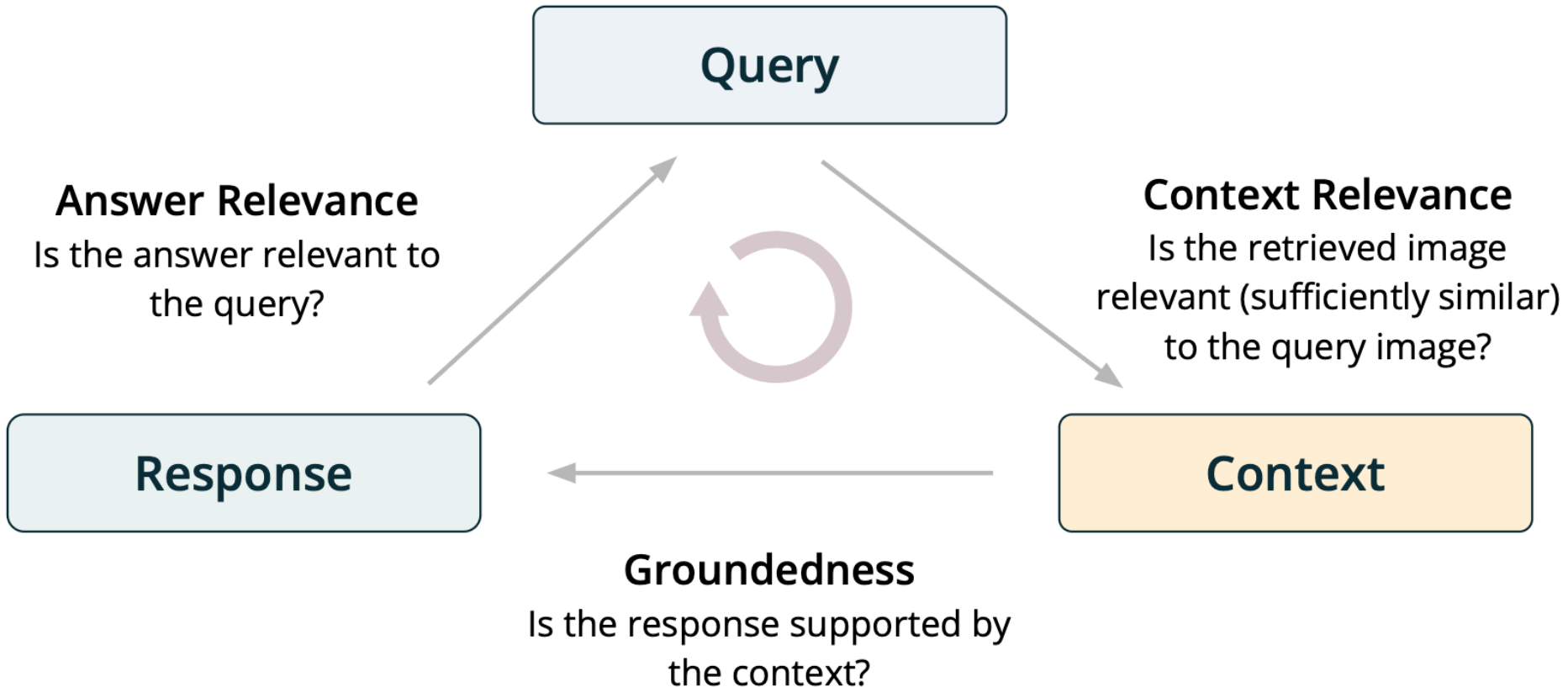
- Whether the **claims** can be supported by retrieved context



The RAG Triad for MRAG?

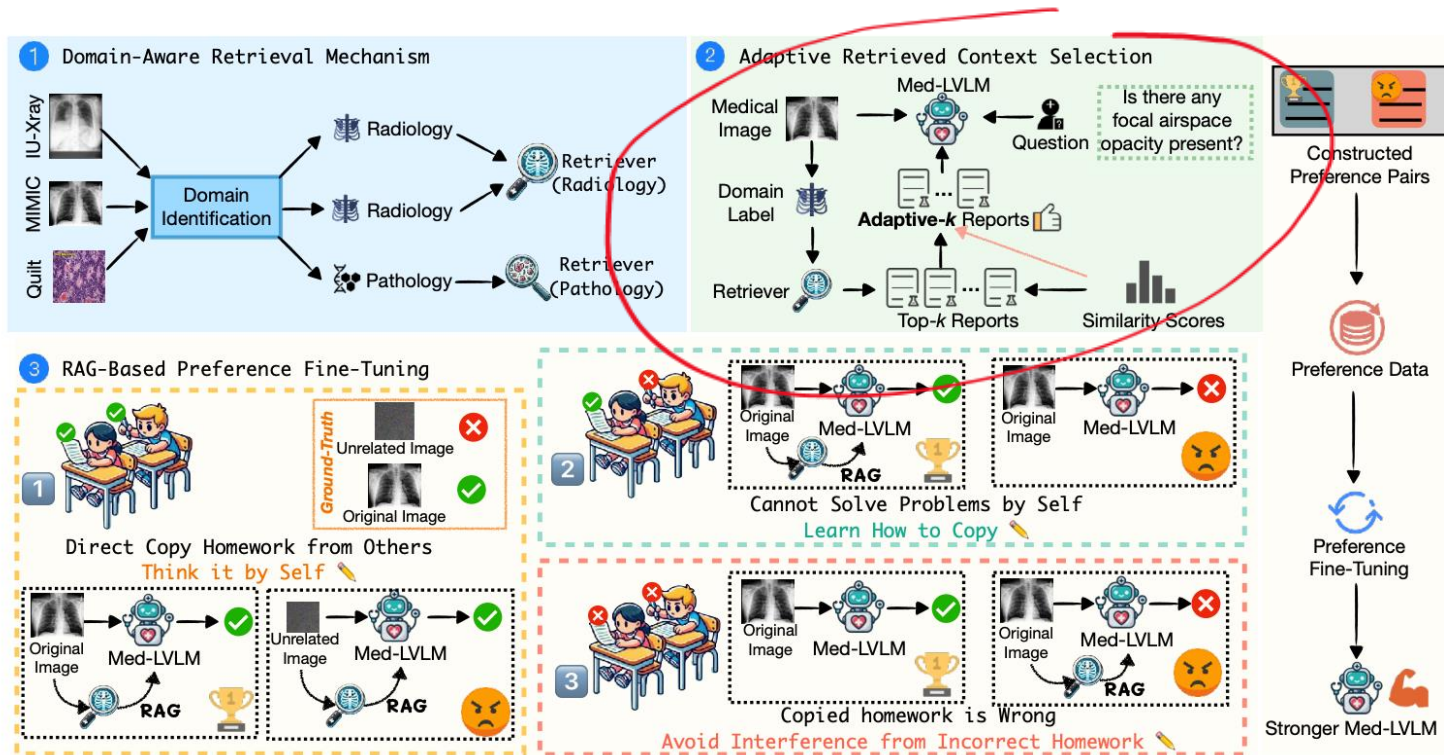
The RAG Triad

Can we use the same LLM-as-a-judge strategy when context is multimodal?



Case Study: The RAG Triad for MRAG?

- Suppose that we have a multimodal RAG system for radiology report generation



Published as a conference paper at ICLR 2025

MMED-RAG: VERSATILE MULTIMODAL RAG SYSTEM FOR MEDICAL VISION LANGUAGE MODELS

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ABSTRACT

Artificial Intelligence (AI) has demonstrated significant potential in healthcare, particularly in disease diagnosis and treatment planning. Recent progress in Medical Large Vision-Language Models (Med-LVLMs) has opened up new possibilities for interactive diagnostic tools. However, these models often suffer from factual hallucination, which can lead to incorrect diagnoses. Fine-tuning and retrieval-augmented generation (RAG) have emerged as methods to address these issues. However, the amount of high-quality data and distribution shifts between training data and deployment data limit the application of fine-tuning methods. Although RAG is lightweight and effective, existing RAG-based approaches are not sufficiently general to different medical domains and can potentially cause misalignment issues, both between modalities and between the model and the ground truth. In this paper, we propose a versatile multimodal RAG system, MMed-RAG, designed to enhance the factuality of Med-LVLMs. Our approach introduces a domain-aware retrieval mechanism, an adaptive retrieved contexts selection, and

Case Study: The RAG Triad for MRAG?

1. Scrape images (x-rays) and paired annotations (diagnosis)



Acute Pulmonary Edema

2. Generate image embeddings and load into vector database with metadata (diagnosis)

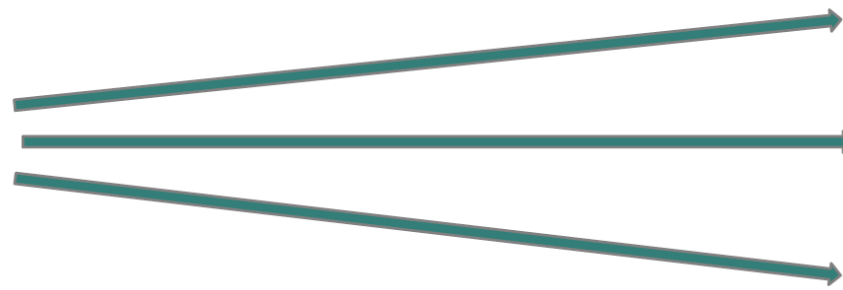
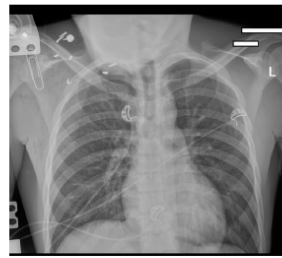
[0.42, 0.35, 0.87, ...]

Acute Pulmonary Edema



Case Study: The RAG Triad for MRAG?

3. Take user image, generate its embeddings
4. Retrieve similar image embeddings and their attached metadata (diagnosis)



Diagnosis A

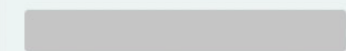
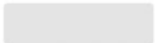
Diagnosis B

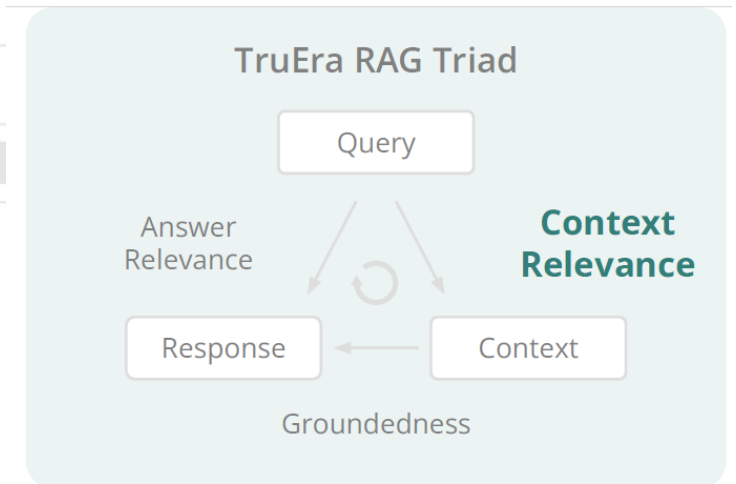
Diagnosis A

5. Generate new diagnosis from shared diagnoses between retrieved images



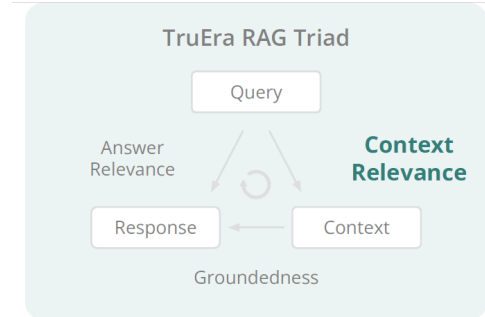
Case Study: The RAG Triad for MRAG?

▼ XRay_Insight.diagnose	2526 ms	
▼ XRay_Insight.generate_embedding	769 ms	
XRay_Insight.preprocess_image	29 ms	
XRay_Insight.compute_embedding	359 ms	
XRay_Insight.reshape_embedding	27 ms	
XRay_Insight.pad_embedding	27 ms	
XRay_Insight.flatten_embedding	26 ms	
XRay_Insight.retrieve_cases	580 ms	
XRay_Insight.generate_diagnosis	994 ms	



Case Study: The RAG Triad for MRAG?

- We may start with evaluating the context relevance from the **text** perspective

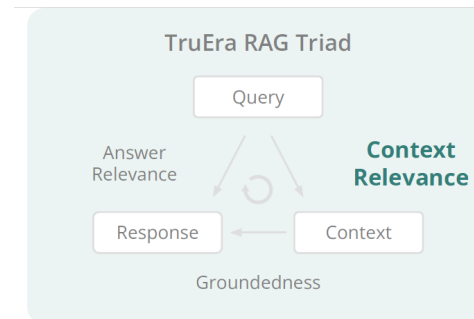


	question	statement	result	reason
0	What is the diagnosis?	Diffuse panbronchiolitis. Normal chest x-ray.	1.0	Criteria: Relevance of the statement to the diagnosis question. S
1	What is the diagnosis?	Pericardial effusion. The cardiac silhouette is enlarged. There is minimal left pleural e	1.0	Criteria: The statement provides relevant information about the
2	What is the diagnosis?	Malpositioned PICC line. Left-sided PICC line with the tip well within the r		Criteria: The statement provides relevant information about the diagnosis. Supporting Evidence: The statement clearly states that the diagnosis is a malpositioned PICC line. It also provides additional information about the specific location of the PICC line tip within the right atrium.

Challenge: *Text Query is **not enough** to know if the diagnosis is relevant*

Case Study: The RAG Triad for MRAG?

- Then we evaluate the context relevance from the **image** perspective...



Input

Input [[Select.RecordInput](#)]

test_set/case_7.png

Response

Response [[Select.RecordOutput](#)]

Radiologic Diagnosis: Hydropneumothorax with mediastinal shift and constrictive pericarditis.



Feedback Metadata

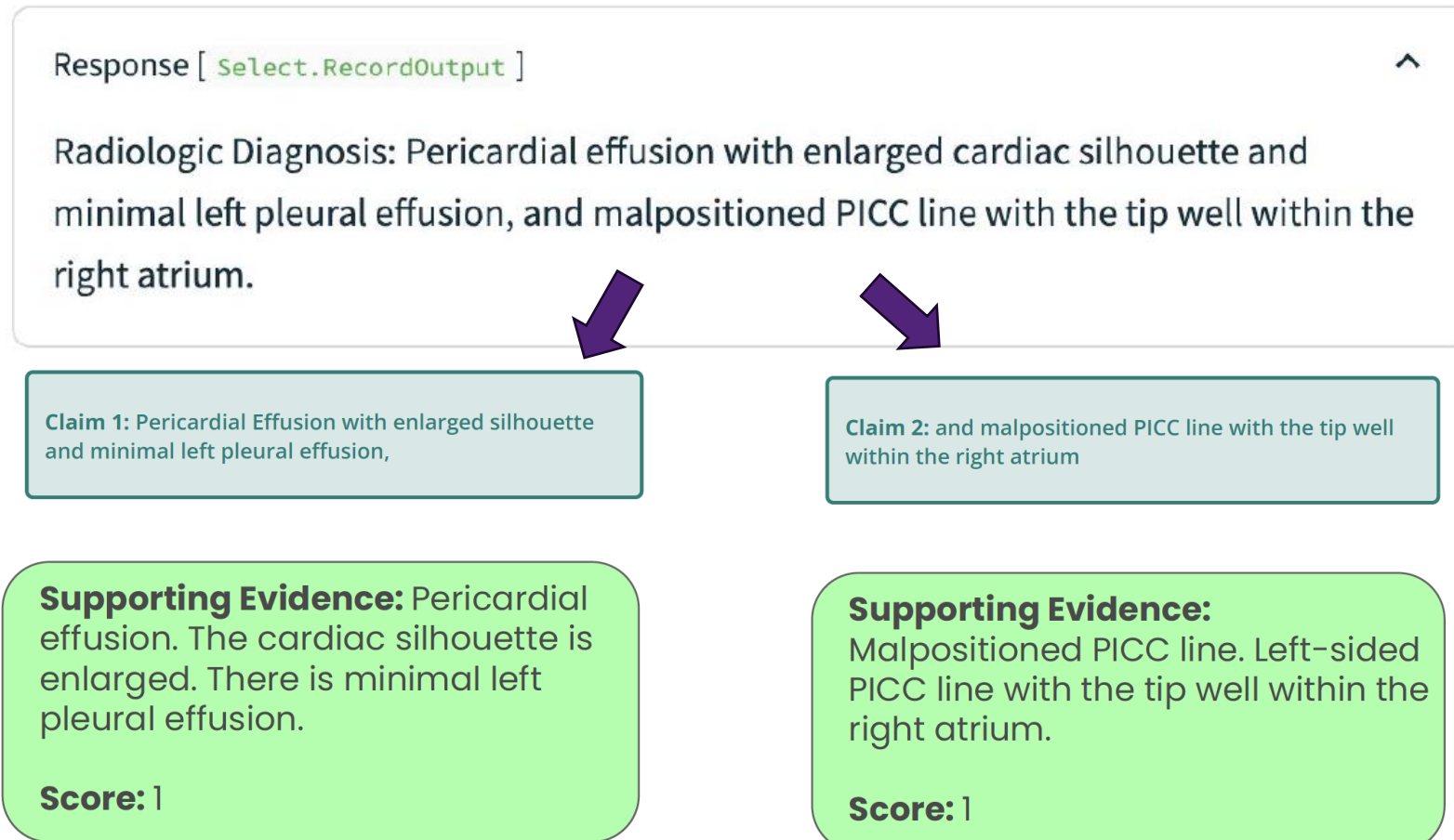
Cosine Similarity = 0.6907378595

	vector1					vector2					result	
0	-0.2632743716	0.5368689895	-2.5768270493	2.7215635777	-1.983833789	0.41831094	2.84881115	-3.43250275	5.26508665	-3.84578967	-1.2986	0.69
1	-0.2632743716	0.5368689895	-2.5768270493	2.7215635777	-1.983833789	2.17195964	2.48818779	-2.89243889	5.50596428	-4.89113426	-1.0852	0.69
2	-0.2632743716	0.5368689895	-2.5768270493	2.7215635777	-1.983833789	1.11391091	3.47702551	-2.86771536	5.30943346	-3.37935257	-0.9293	0.69

**High cosine similarity
But wrong context..**

Case Study: The RAG Triad for MRAG?

- Groundedness / Faithfulness

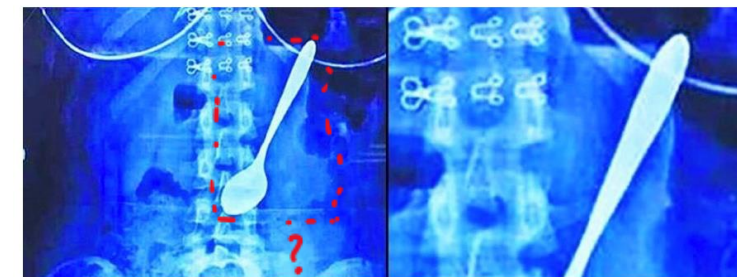
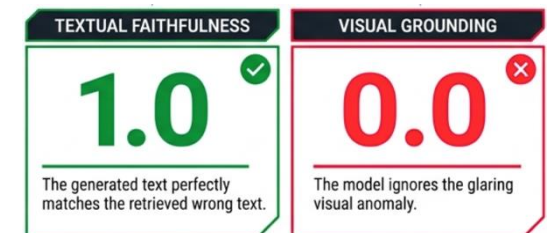
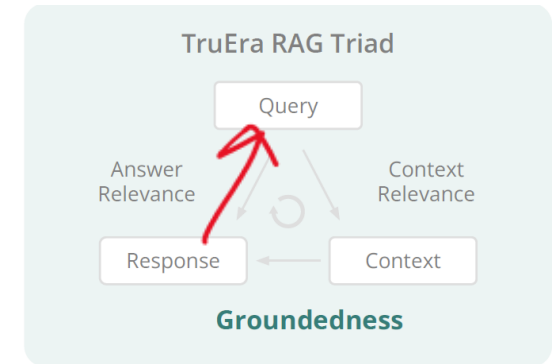
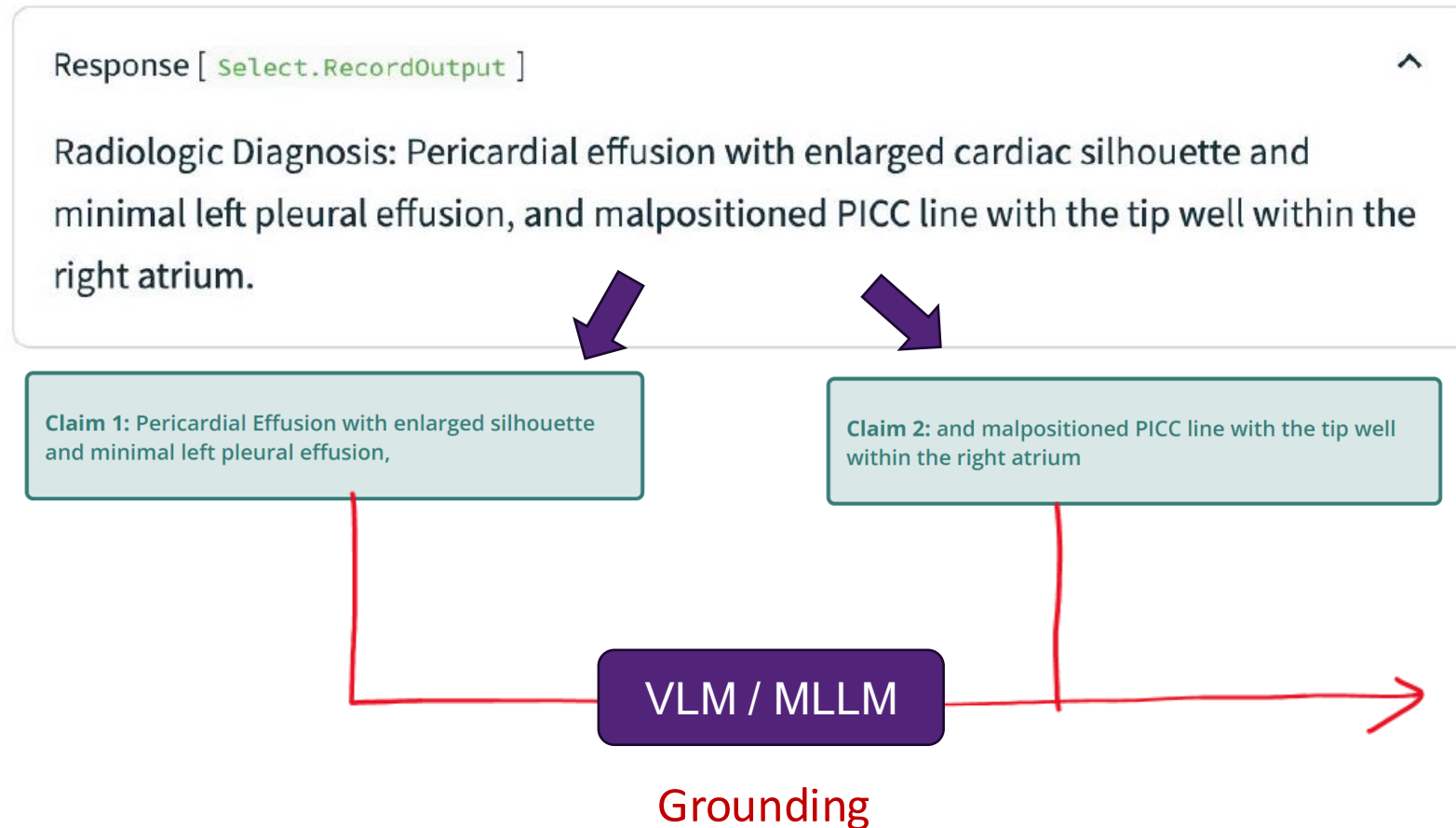


Wrong context misleads..

Text-centric eval all fail in this case...

Case Study: The RAG Triad for MRAG?

- How to address? -> **Query Grounding**



Non-exist!

Defining Hallucinations in Multimodal RAG

MLLM hallucination

- Textual answer describes visual input incorrectly.
- Typical unit: image/video → generated answer.
- Common types: nonexistent object, wrong attribute, wrong relation, wrong event.
- The input visual signal is the main grounding source.

Image / Video



MLLM

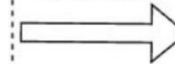


Answer

Prompt: Please provide a detailed description for items in the fridge.



LVLM



Response: In the image, we see the contents of a refrigerator shelf. Starting from the left, there's a large transparent plastic bag containing what appears to be carrots. Next to the carrots is a bottle of **juices** with a red cap. There are **two boxes of strawberries**. The box filled with strawberries is located to **the right** of the bag of carrots.....



*Hallucinated response visualized. Not part of the response

Defining Hallucinations in Multimodal RAG

Object existence

- The answer mentions an entity not present in the image, video, source.

Attribute

- The entity exists, but color, count, value, state, or label is wrong.

Relation

- Entities exist, but spatial, causal, or **logical** relation is wrong.

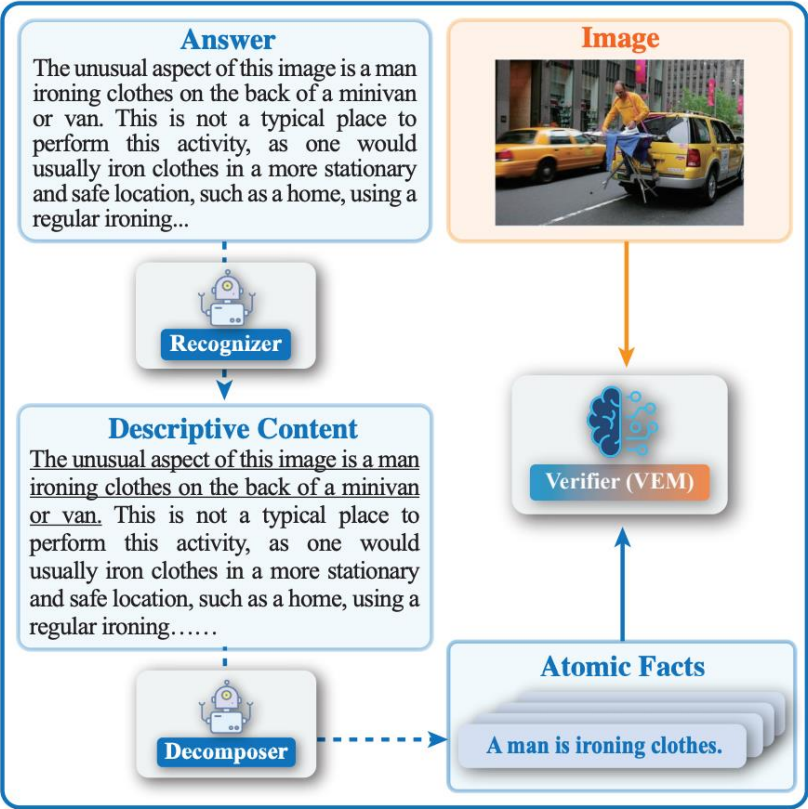
Event / temporal

- Video/audio sequence, action order, or time segment is misread.

Use this taxonomy first; then add RAG-specific failure modes.

Defining Hallucinations in Multimodal RAG

FaithScore: Quantify Hallucinations



Prompt:
Can you elaborate on the elements of the picture provided?

Image:

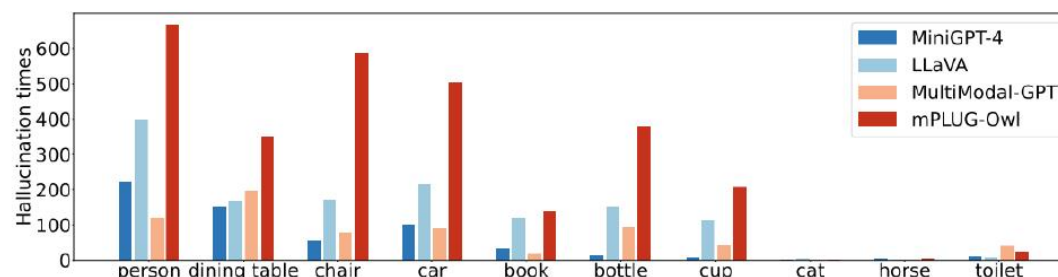
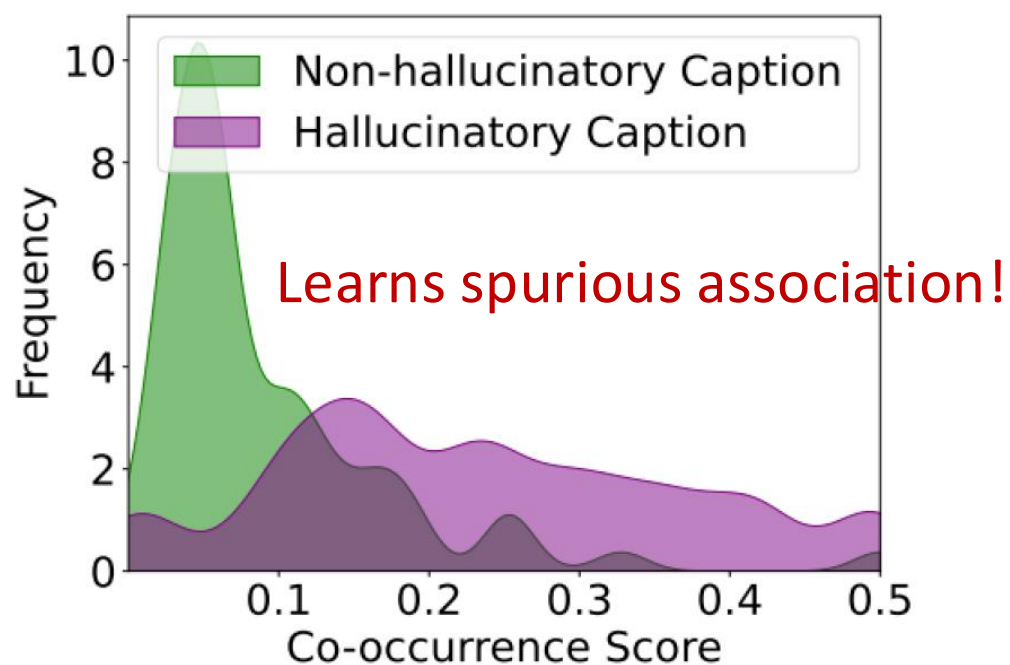
Answer:
The image features a desk with a newspaper, a pair of sunglasses, and a book. The desk is cluttered with various items, including a pen, a pencil, and a notebook. The newspaper is spread out on the desk, and the sunglasses are placed on top of it. The book is also present on the desk, possibly indicating that the person using the desk is reading or studying. The overall scene suggests a workspace or study area where the person has various items and resources available for their tasks

Atomic Facts	
- There is a desk.	✓
- There is a newspaper.	✓
- The newspaper is on the desk.	✓
- There is a pair of sunglasses.	✓
- There is a book.	✓
- There are various items.	✓
- The items are cluttered on the desk.	✓
- The desk is cluttered.	✓
- There is a pen.	✗
- There is a pencil.	✗
- There is a notebook.	✓
- The newspaper is spread out on the desk.	✓
- The sunglasses are placed on top of newspaper.	✓
- The book is on the desk.	✓
FaithScore: 86%	

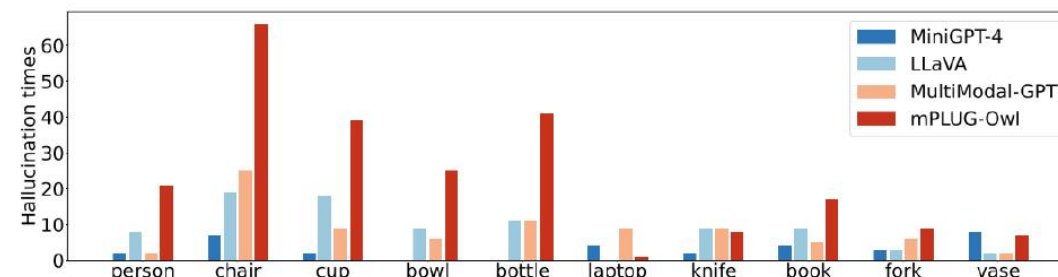
Defining Hallucinations in Multimodal RAG

Why do hallucination exist? (Statistical Bias)

- LVLM/MLLM possess an intrinsic tendency to memorize training data.
- The nouns (e.g., objects) distribution in the training dataset has strong effects on the behavior of the the model.



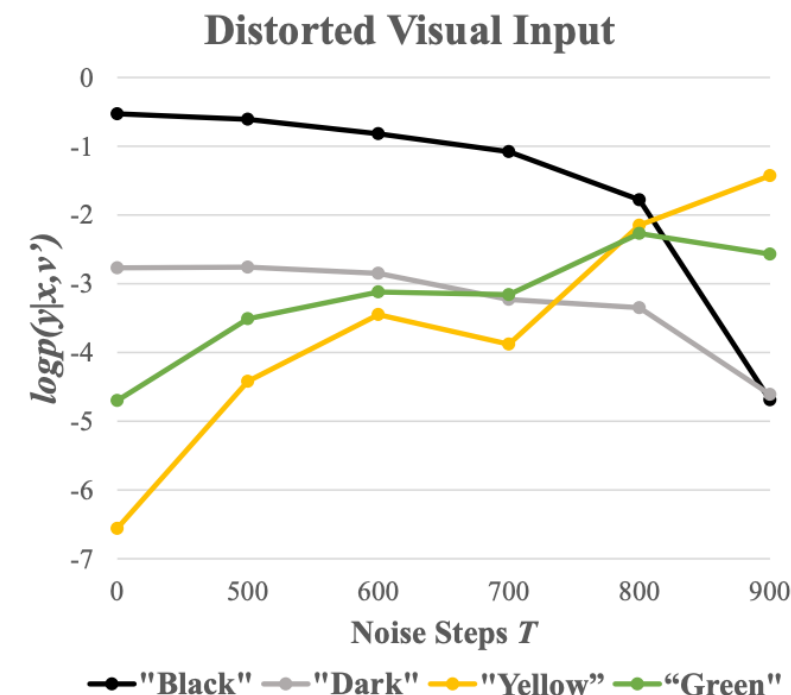
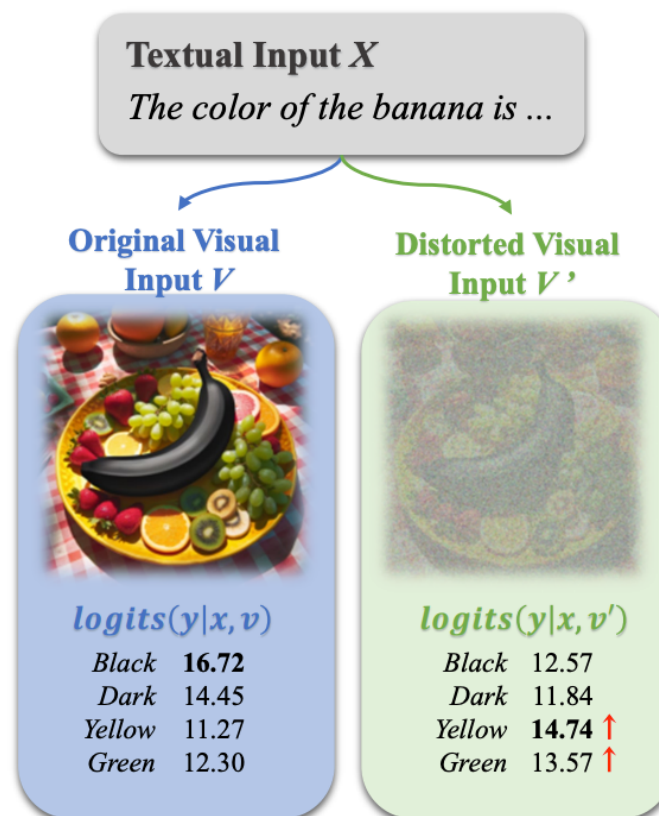
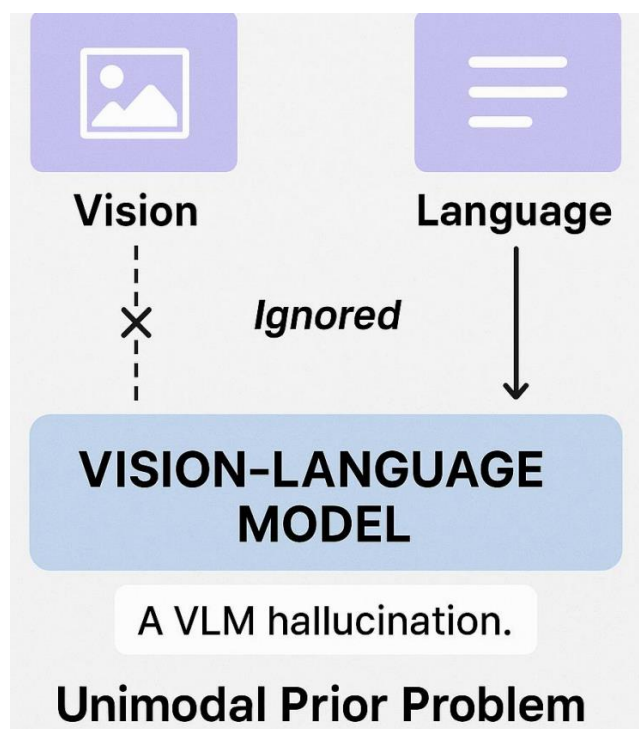
(a) Hallucination times of top ten frequently appearing objects, whose frequencies decrease from right to left.



(b) Hallucination times of top ten objects co-occurring with "dining table", whose frequencies decrease from right to left.

Defining Hallucinations in Multimodal RAG

Why do hallucination exist? (Visual Uncertainty)



over reliance on language knowledge
when **visual evidence is weak**.

Apply Gaussian noise

VCD (Visual Contrastive Decoding)

What is VCD?

- A **training-free** method to mitigate object hallucination.
- Contrasts **output distribution** from original and distorted inputs.
- Reduces over reliance on statistical biases and unimodal priors.

(self-correcting)

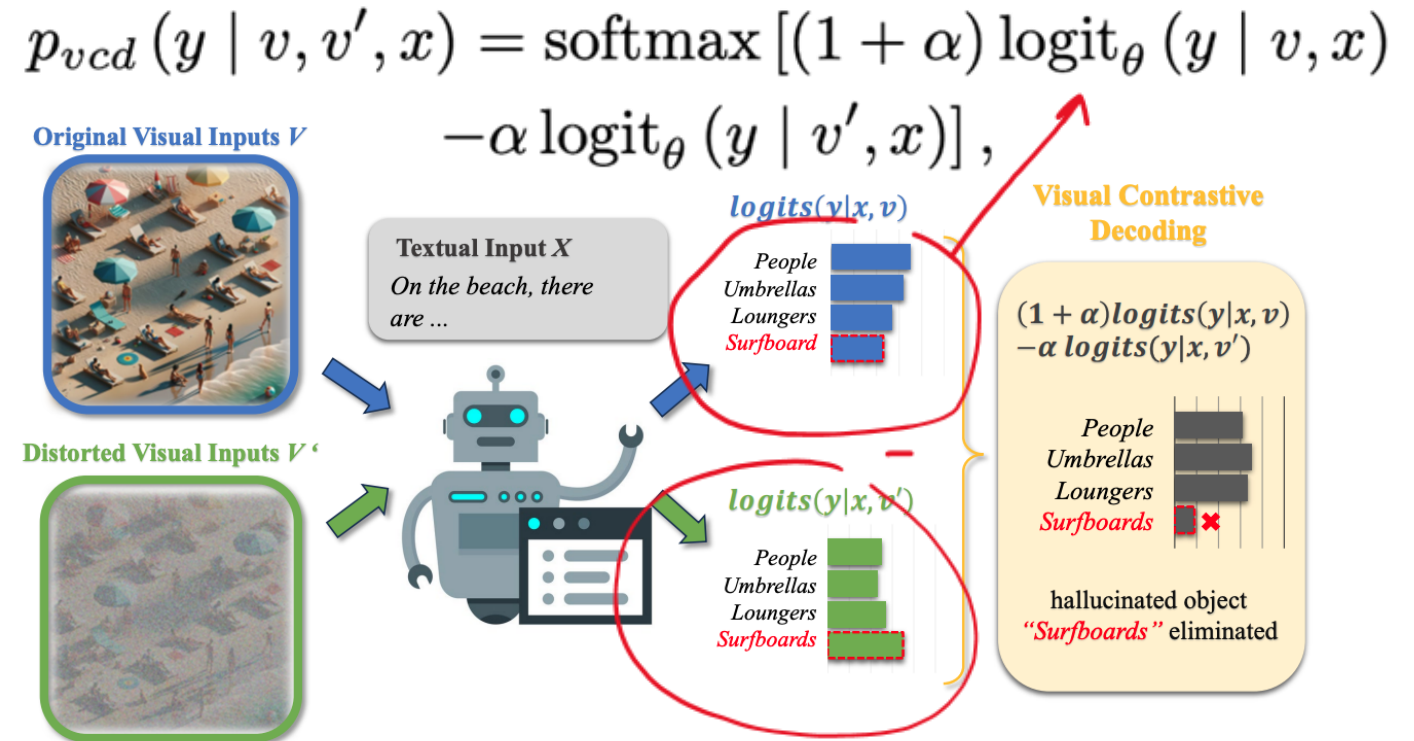


Figure 1. An illustration of Visual Contrastive Decoding. The hallucinated object "Surfboards" is highlighted in red, and it is eliminated during the generative process by contrasting with the output distribution that favors hallucinations.

VCD (Visual Contrastive Decoding)

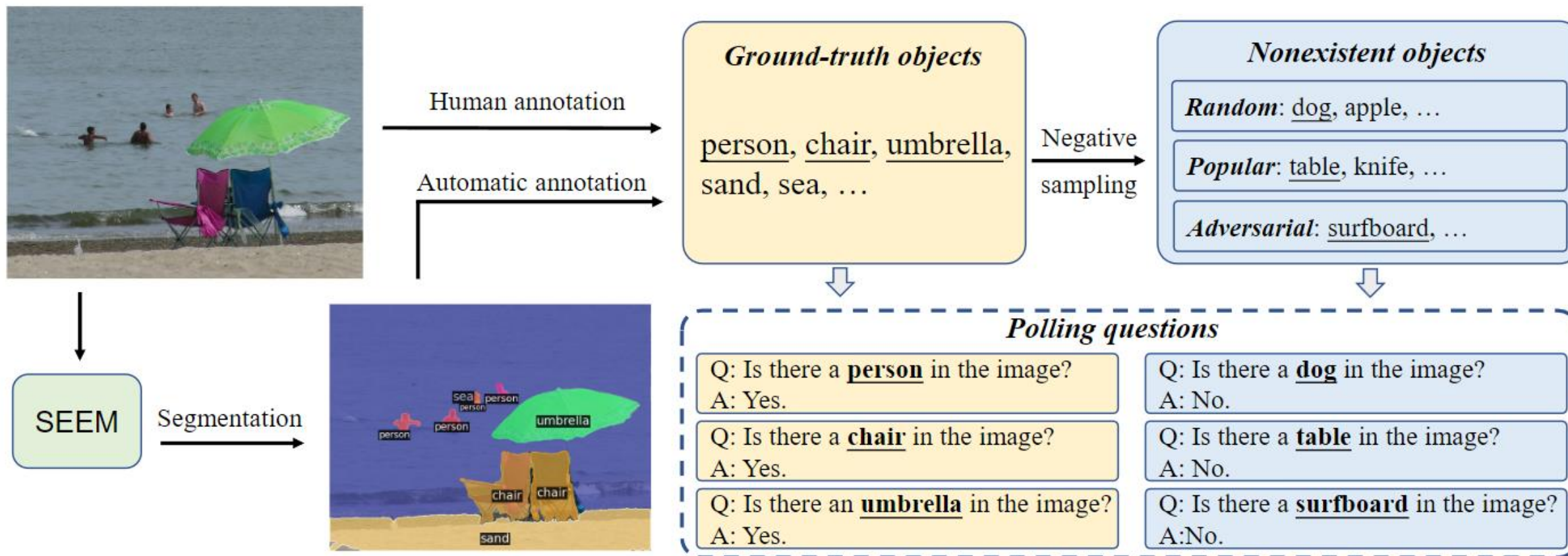
Prompt: Describe this photo in detail.

Regular Decoding: The image features a painting of a brown dog sitting in a chair, with a turkey feather in its mouth, giving it a unique, conspicuous appearance. The dog's expression is one of mischief as it looks directly into the camera. The scene is set in a room with a **dining table** and a **fork** next to the **table**. A rock, likely from the surrounding landscape, can be seen beside the **table**. A **person's** face is visible near the bottom left corner of the painting, partially covered by the dog's furry coat.



Visual Contrastive Decoding: The image features a unique and amusing painting of a dog with a large brush mustache, giving it an artistic human-like appearance. The dog is sitting on a couch, with a red background behind it, adding to the artistic flair of the painting. The couch has an orange and green tint to it. In addition to the dog's human-like mustache, the painting also includes a monochromatic style with the background, giving the painting a striking visual appeal. The dog appears to be wearing a hat, further emphasizing the artistic interpretation of the scene. Overall, the painting captures the dog's whimsical and charming personality.

VCD (Visual Contrastive Decoding)



Benchmarks

Benchmark	Primary Focus	Task	Evaluation
POPE	Object Hallucination (existence)	Binary Q&A (Yes/No)	Quantitative (F1, Accuracy)

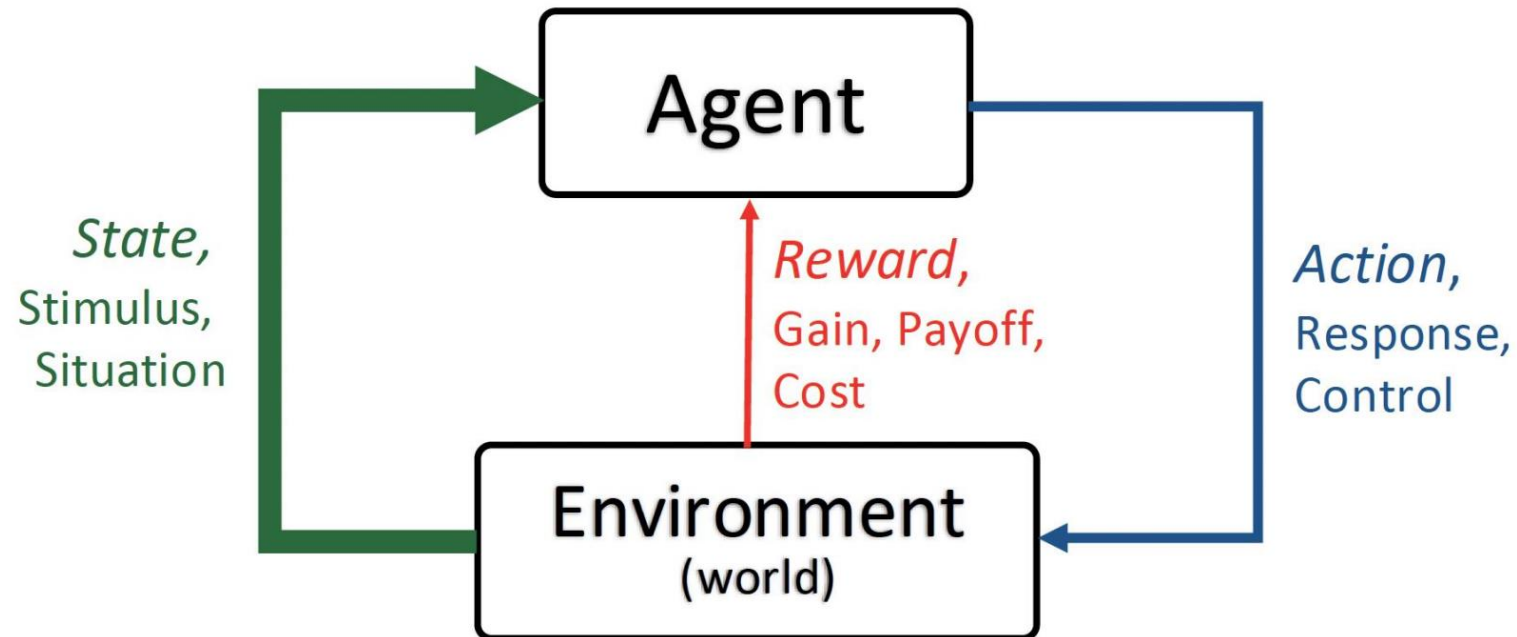
VCD (Visual Contrastive Decoding)

Can we do better beyond training-free approaches?

Dataset	Setting	Model	Decoding	Accuracy↑	Precision	Recall	F1 Score↑
MSCOCO	<i>Random</i>	LLaVA1.5	Regular	83.29 _(±0.35)	92.13 _(±0.54)	72.80 _(±0.57)	81.33 _(±0.41)
			VCD	87.73 _(±0.40)	91.42 _(±0.55)	83.28 _(±0.42)	87.16 _(±0.41)
		Qwen-VL	Regular	84.73 _(±0.36)	95.61 _(±0.45)	72.81 _(±0.38)	82.67 _(±0.41)
			VCD	88.63 _(±0.10)	94.64 _(±0.25)	81.91 _(±0.19)	87.81 _(±0.11)
		InstructBLIP	Regular	80.71 _(±0.73)	81.67 _(±0.67)	79.19 _(±1.14)	80.41 _(±0.80)
			VCD	84.53 _(±0.38)	88.55 _(±0.54)	79.32 _(±0.44)	83.68 _(±0.40)
	<i>Popular</i>	LLaVA1.5	Regular	81.88 _(±0.48)	88.93 _(±0.60)	72.80 _(±0.57)	80.06 _(±0.05)
			VCD	85.38 _(±0.38)	86.92 _(±0.53)	83.28 _(±0.42)	85.06 _(±0.37)
		Qwen-VL	Regular	84.13 _(±0.18)	94.31 _(±0.43)	72.64 _(±0.45)	82.06 _(±0.23)
			VCD	87.12 _(±0.07)	91.49 _(±0.10)	81.85 _(±0.19)	86.40 _(±0.09)
		InstructBLIP	Regular	78.22 _(±0.84)	77.87 _(±1.03)	78.85 _(±0.52)	78.36 _(±0.76)
			VCD	81.47 _(±0.42)	82.89 _(±0.64)	79.32 _(±0.44)	81.07 _(±0.39)
	<i>Adversarial</i>	LLaVA1.5	Regular	78.96 _(±0.52)	83.06 _(±0.58)	72.75 _(±0.59)	77.57 _(±0.57)
			VCD	80.88 _(±0.33)	79.45 _(±0.29)	83.29 _(±0.43)	81.33 _(±0.34)
		Qwen-VL	Regular	82.26 _(±0.30)	89.97 _(±0.33)	72.61 _(±0.50)	80.37 _(±0.37)
			VCD	84.26 _(±0.39)	85.84 _(±0.45)	82.05 _(±0.39)	83.90 _(±0.39)
		InstructBLIP	Regular	75.84 _(±0.45)	74.30 _(±0.63)	79.03 _(±0.68)	76.59 _(±0.40)
			VCD	79.56 _(±0.41)	79.67 _(±0.59)	79.39 _(±0.50)	79.52 _(±0.38)

Mitigate Hallucinations via RL

Teach the model to prefer not to hallucinate with reinforcement learning (RL)



- **Environment** may be unknown, non-linear, stochastic and complex.
- **Agent** learns a **policy** to map states of the environments to actions.
 - Seeking to maximize cumulative reward in the long run.

Mitigate Hallucinations via RL

$$\hat{r}(x_T, x_I, y) = \beta \log \frac{\pi_{\theta}(y_{pos} | [x_T, x_I])}{\pi_{ref}(y_{pos} | [x_T, x_I])}$$

Teach the model to prefer not to hallucinate with reinforcement learning (RL)

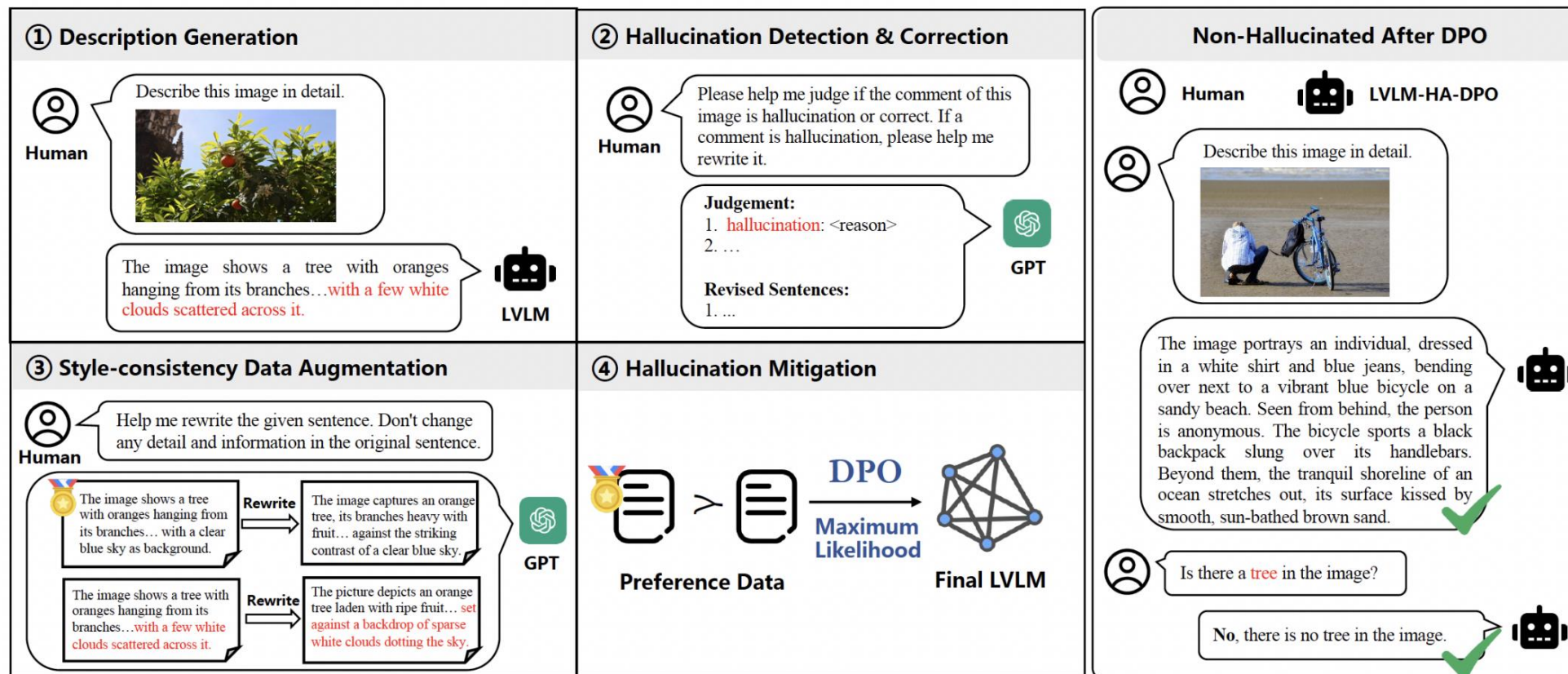


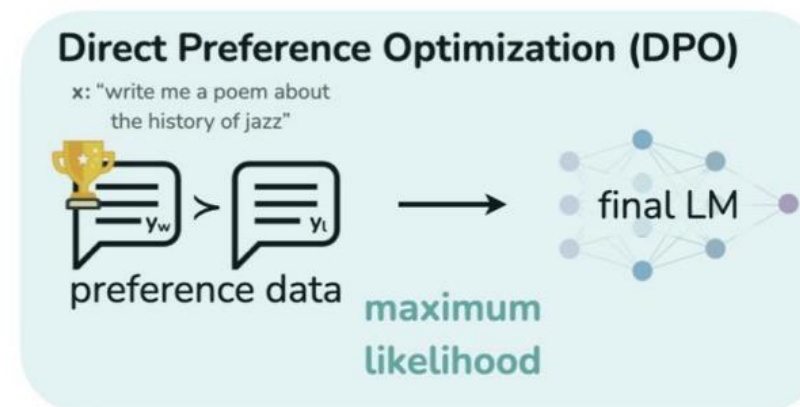
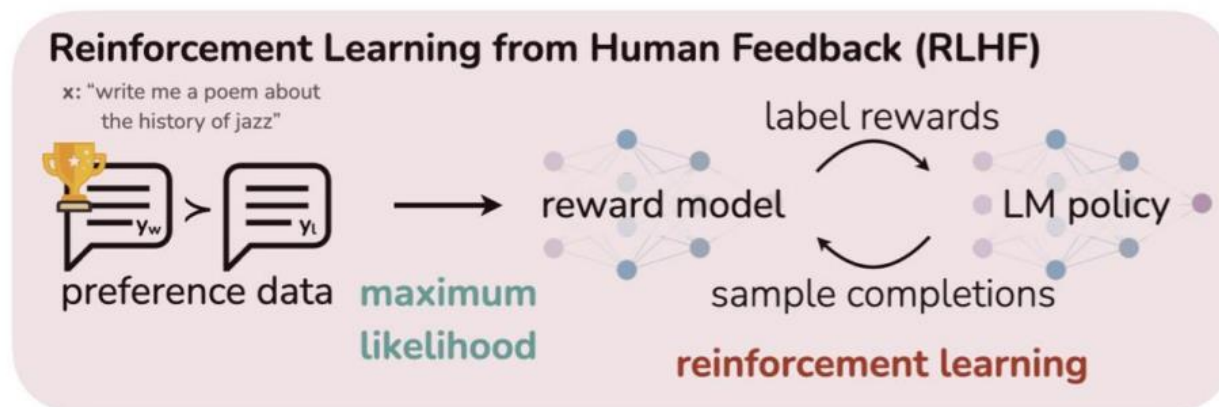
Figure 2. Our proposed hallucination mitigation process involves four steps: (1) **Description Generation**, where the LVLM is tasked with a detailed image description; (2) **Hallucination Detection and Correction**, GPT-4 identifies and corrects hallucinations in model responses using rich annotations; (3) **Style-consistency Data Augmentation**, GPT-4 rewrites samples to maintain style consistency; and (4) **Hallucination Mitigation**, style-consistent data is gathered for DPO training.

Mitigate Hallucinations via RL

Teach the model to prefer not to hallucinate with reinforcement learning (RL)

Direct Preference Optimization

- DPO skips explicit reward modeling and RL
- models are trained to assign higher likelihood to preferred outputs over rejected ones, directly optimizing next-token orderings based on preference data
- DPO is efficient and popular for preference learning without RL overhead.



Mitigate Hallucinations via RL

Teach the model to prefer not to hallucinate with reinforcement learning (RL)

POPE	Model	HA-DPO	Accuracy	Precision	F1 Score	Yes Ratio (%)
Random	MiniGPT-4-LLama2-7B [36]	X	51.13	50.57	67.13	98.66
		✓	86.13	92.81	84.96	42.20
	InstructBLIP-13B [5]	X	88.70	85.03	89.26	55.23
		✓	89.83	93.07	89.43	46.23
	LLaVA-1.5-7B [13]	X	89.60	88.77	89.70	51.06
		✓	90.53	92.99	90.25	47.13
Popular	MiniGPT-4-LLama2-7B [36]	X	51.46	50.74	67.72	98.06
		✓	79.50	80.20	79.25	48.83
	InstructBLIP-13B [5]	X	81.36	75.06	83.44	62.56
		✓	85.76	85.55	85.80	50.03
	LLaVA-1.5-7B [13]	X	86.20	83.23	86.79	54.46
		✓	87.90	88.07	87.81	49.76
Adversarial	MiniGPT-4-LLama2-7B [36]	X	51.26	50.64	67.16	98.40
		✓	75.66	74.36	76.29	52.66
	InstructBLIP-13B [5]	X	74.50	67.64	78.64	69.43
		✓	80.70	77.72	81.68	55.36
	LLaVA-1.5-7B [13]	X	79.76	74.43	81.75	60.90
		✓	81.46	77.99	82.54	56.20

Defining Hallucinations in Multimodal RAG

Query-grounding

- Misreads the user-provided image/chart/video before retrieval

Retrieval hallucination

- Answer fills in facts because required evidence was not retrieved

Grounding hallucination

- Evidence is retrieved but visually misperceived

Reasoning hallucination

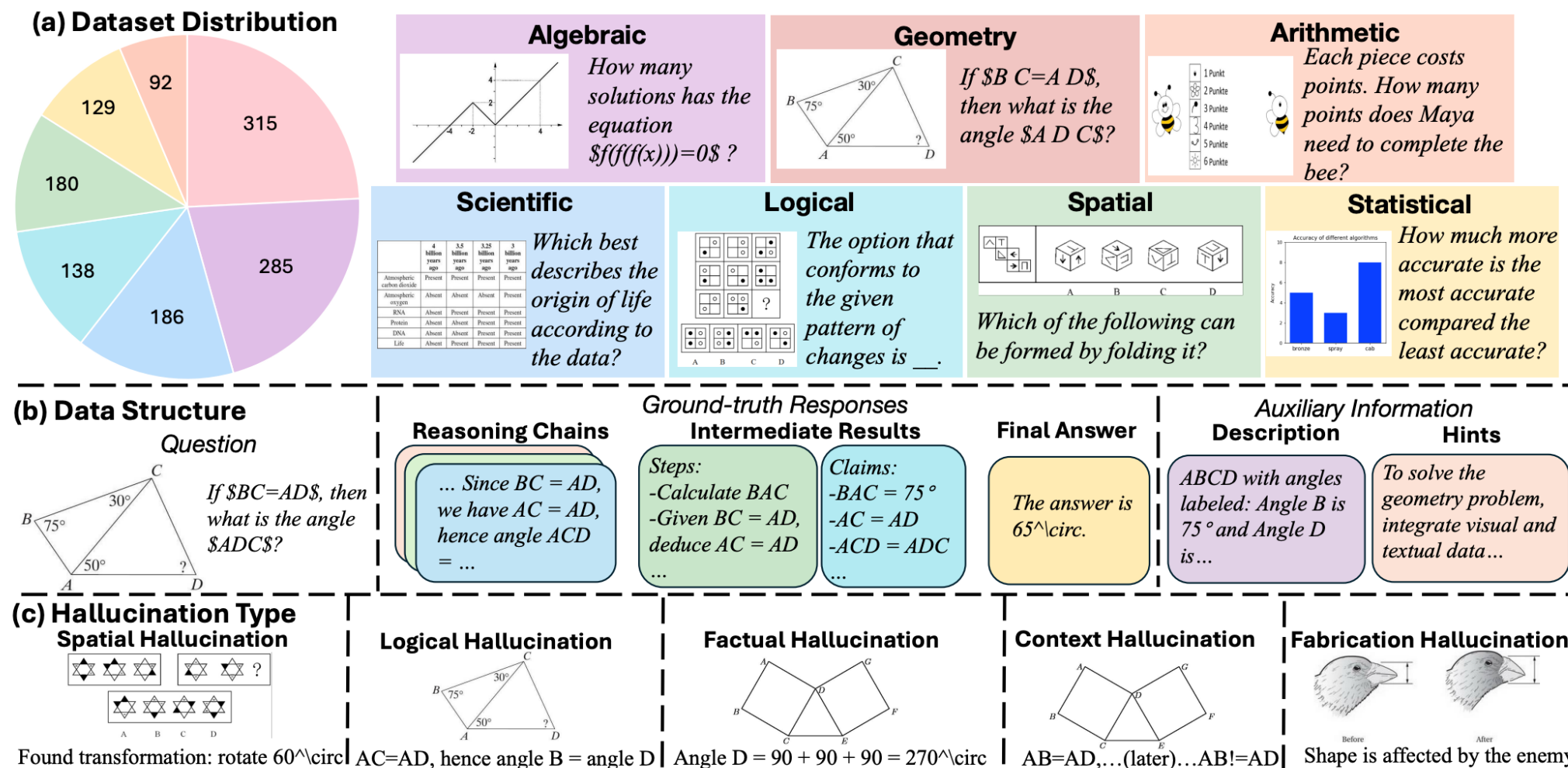
- Combines facts from multiple sources into a false conclusion

Citation hallucination

- Claim may be true, but cited source does not support it

M-RAG hallucination

MiRAGE: Assessing Hallucination in Multimodal Reasoning Chains of MLLM



MiRAGE: Assessing Hallucination in Multimodal Reasoning Chains of MLLM

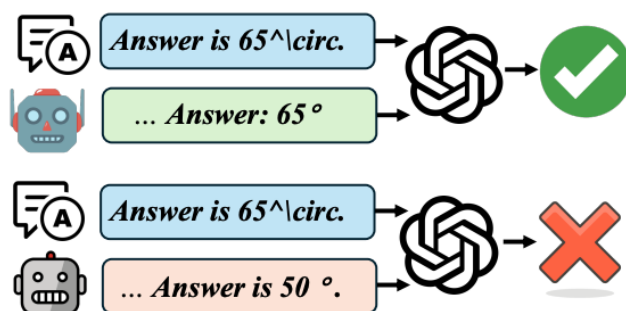
(a) Dataset Construction



(b) Dataset Evaluation

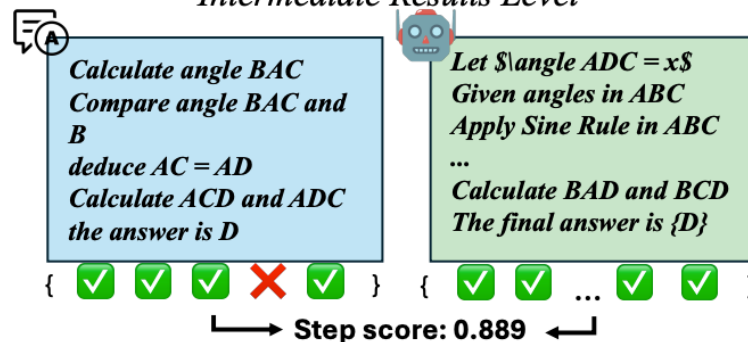
Accuracy Assessment

Final Answer Level



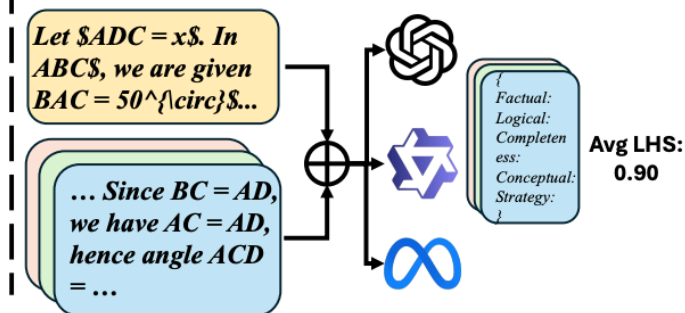
Factuality Assessment

Intermediate Results Level

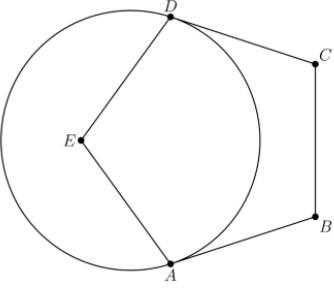


LLMs Hallucination Score

Reasoning Chain Level



MiRAGE: Assessing Hallucination in Multimodal Reasoning Chains of MLLM



Original Chain:

...

The sum of angles around a point is (360°) . The angles around point (E) , inside the circle, are $(\angle ADE + \angle DEA + \angle EAD = 360^\circ)$.

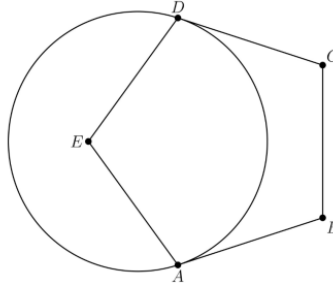
Since both $(\angle ADE)$ and $(\angle EAD)$ are (90°) , the angle $(\angle DEA)$ is:

...

...What is the measure of minor arc AD ?



Hence, the measure of minor arc AD is 180° .



Manually Corrected Chain:

...

In a regular pentagon, each internal angle is 108° , and the central angle between adjacent vertices is 72° (since $360^\circ/5 = 72^\circ$). The sides $\overline{AB}$ and $\overline{DC}$ are separated by two vertices (B and C). The angle between these sides corresponds to $2 \times 72^\circ = 144^\circ$.

...

...What is the measure of minor arc AD ?



Thus, the measure of minor arc AD is the same as the central angle AOD , which is: 144° .

Figure 6: Qualitative results of manually fixing the reasoning hallucination in the reasoning chain and inference the refined answers. The corrected reasoning chain (right) can guide MLLMs to predict correct answers, and the original reasoning chain with hallucination (left) still results in wrong answers.

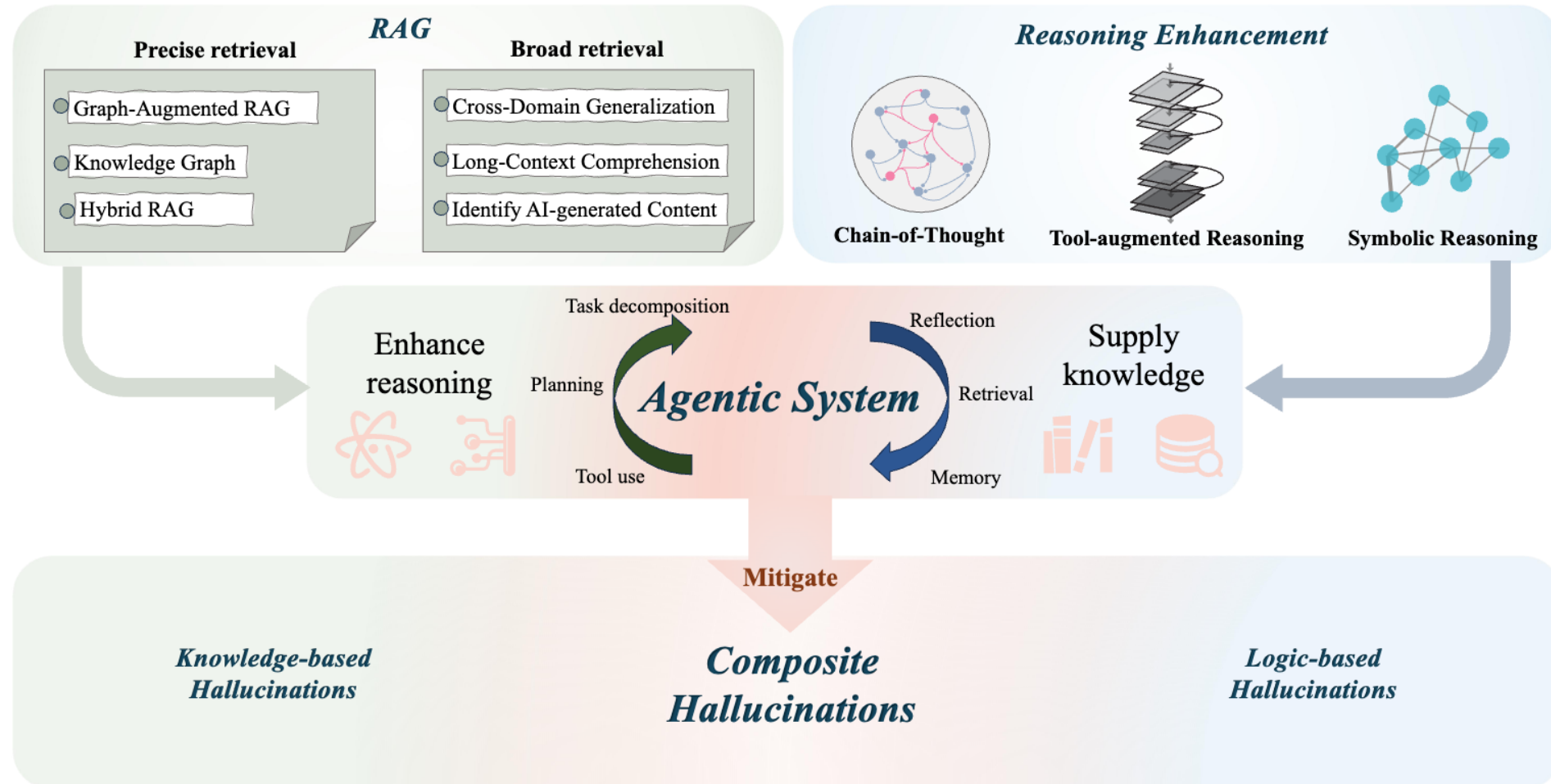
MiRAGE: Assessing Hallucination in Multimodal Reasoning Chains of MLLM

Table 2: Comparison of recent state-of-the-art MLLMs on MIRAGE. Best results are **bolded**.

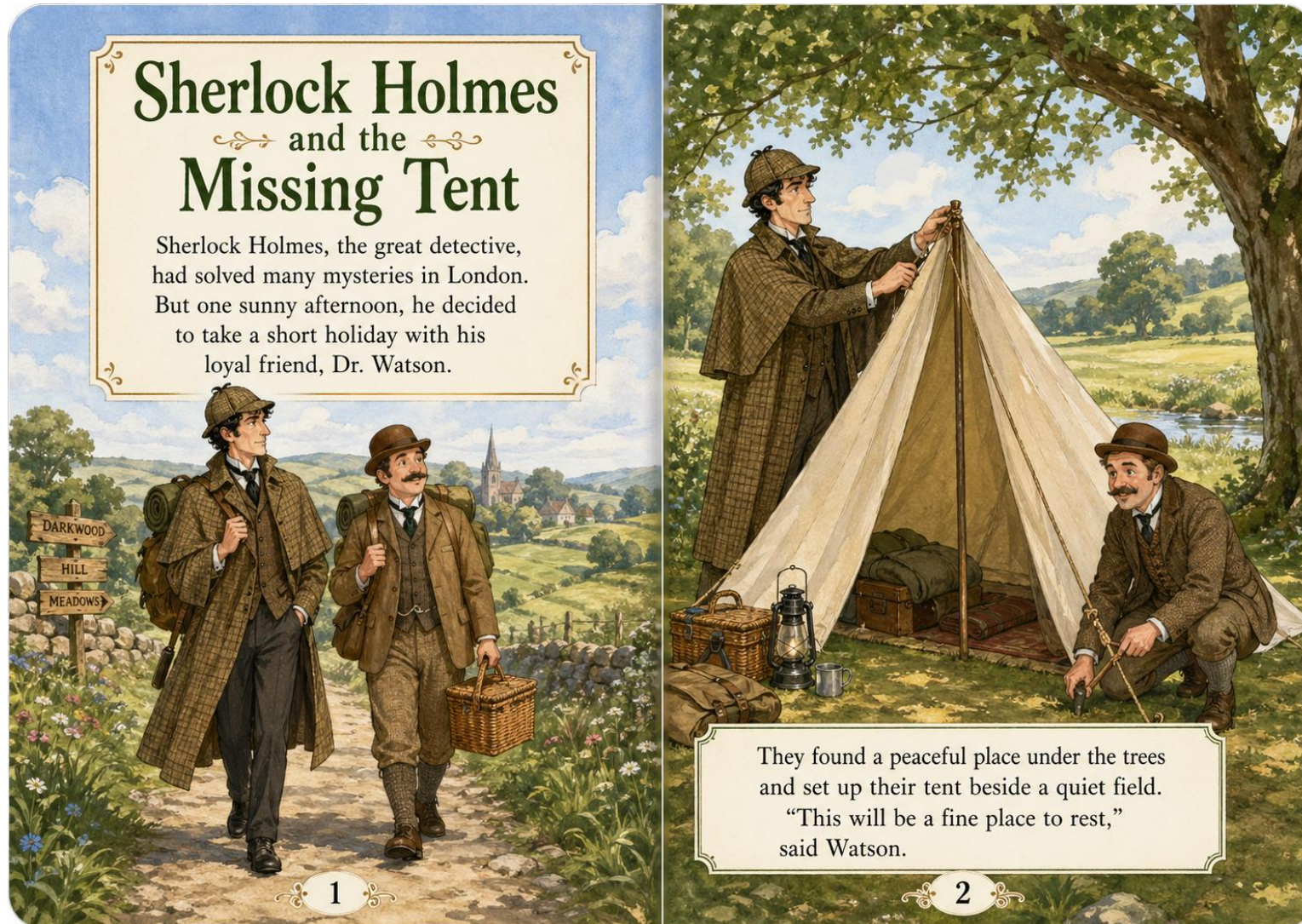
Model	Type	Accuracy $\uparrow$	Factuality		LHS $\uparrow$
			$F_{\text{step}} \uparrow$	$F_{\text{claim}} \uparrow$	
Black-Box MLLMs					
Gemini-2-Flash-Thinking [55]	Reasoning	47.6	51.5	50.7	0.7517$\pm$0.0168
O1 [24]	Reasoning	49.7	41.3	42.7	0.6193 $\pm$ 0.0091
Gemini-2-Flash [55]	General	44.1	47.8	47.4	0.6882 $\pm$ 0.0496
GPT-4o [23]	General	35.0	39.2	40.6	0.6332 $\pm$ 0.0111
Open-sourced $\sim$ 72B MLLMs					
Qwen2.5-VL-72B-Instruct [2]	General	38.8	47.4	44.6	0.7223$\pm$0.0339
InternVL-2.5-78B [5]	General	29.6	39.0	36.6	0.6377 $\pm$ 0.0325
Qwen2-VL-72B-Instruct [60]	General	24.5	29.7	26.2	0.4928 $\pm$ 0.0332
QvQ-72B-Preview [56]	Reasoning	31.0	46.1	45.3	0.5717 $\pm$ 0.0597
Virgo-72B [12]	Reasoning	37.4	47.1	45.0	0.6328 $\pm$ 0.0251
Open-sourced $\sim$ 7B MLLMs					
Qwen2.5-VL-7B-Instruct [2]	General	28.8	34.7	31.7	0.5996 $\pm$ 0.0123
Qwen2-VL-7B-Instruct [2]	General	19.5	21.9	18.6	0.3633 $\pm$ 0.0106
Qwen2.5-VL-7B-Instruct+VIC [85]	Reasoning	26.9	22.8	25.2	0.4478 $\pm$ 0.0177
Qwen2.5-VL-7B-Instruct+Reflection [16]	Reasoning	26.7	40.1	33.4	0.5826 $\pm$ 0.0124
R1-OneVision-7B [5]	Reasoning	22.9	30.7	30.2	0.5098 $\pm$ 0.0099
Mulberry-Qwen2-VL-7B [71]	Reasoning	22.6	29.2	24.4	0.4740 $\pm$ 0.0147
InternVL-2.5-8B [5]	General	20.8	31.9	26.4	0.4838 $\pm$ 0.0156
Llama-3.2-Vision-11B [18]	General	18.7	26.9	22.3	0.4265 $\pm$ 0.0141
Llava-CoT-11B [66]	Reasoning	17.4	26.9	22.4	0.4267 $\pm$ 0.0140
Logos-7B (Ours)	Reasoning	37.1	43.3	38.3	0.6568$\pm$0.0179
Open-sourced $\sim$ 3B MLLMs					
Qwen2.5-VL-3B-Instruct [2]	General	18.8	23.1	18.8	0.3422 $\pm$ 0.0244
Phi-3.5-Instruct [1]	General	12.9	16.6	13.8	0.3181 $\pm$ 0.0161
Logos-3B (Ours)	Reasoning	29.4	38.9	34.5	0.5840$\pm$0.0216

Composite Hallucinations for Agentic RAG

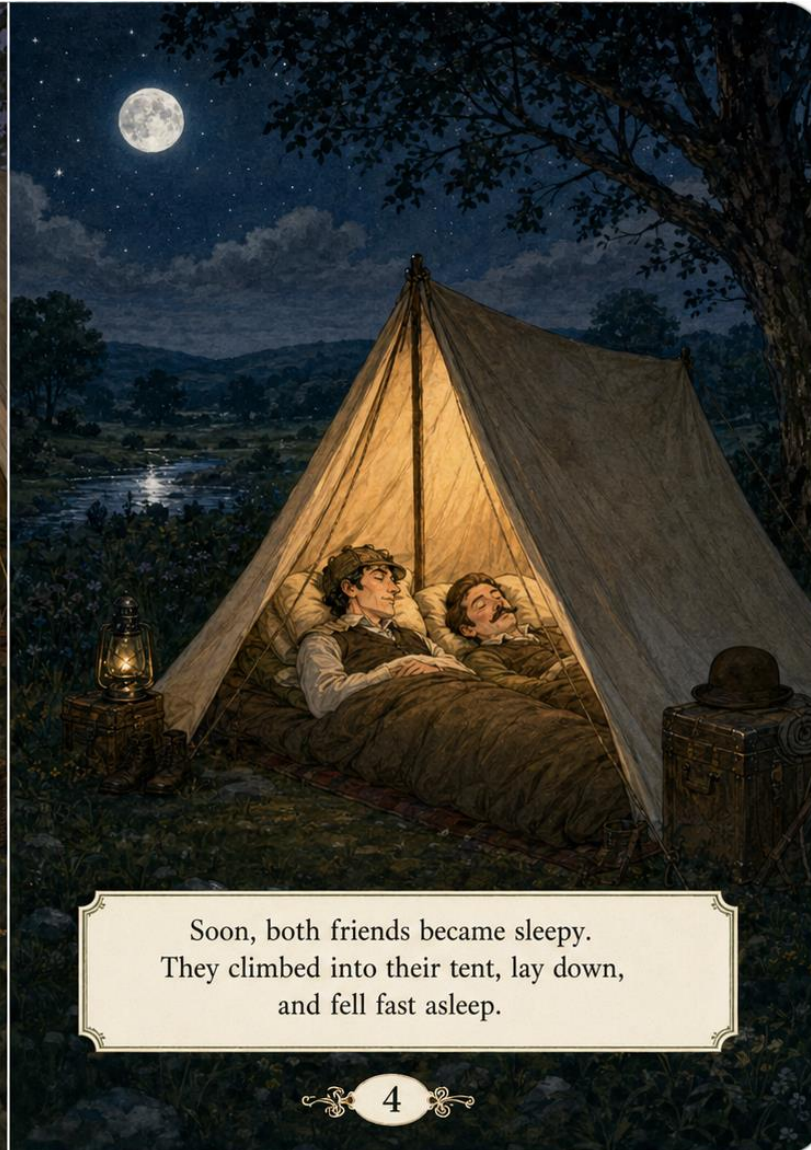
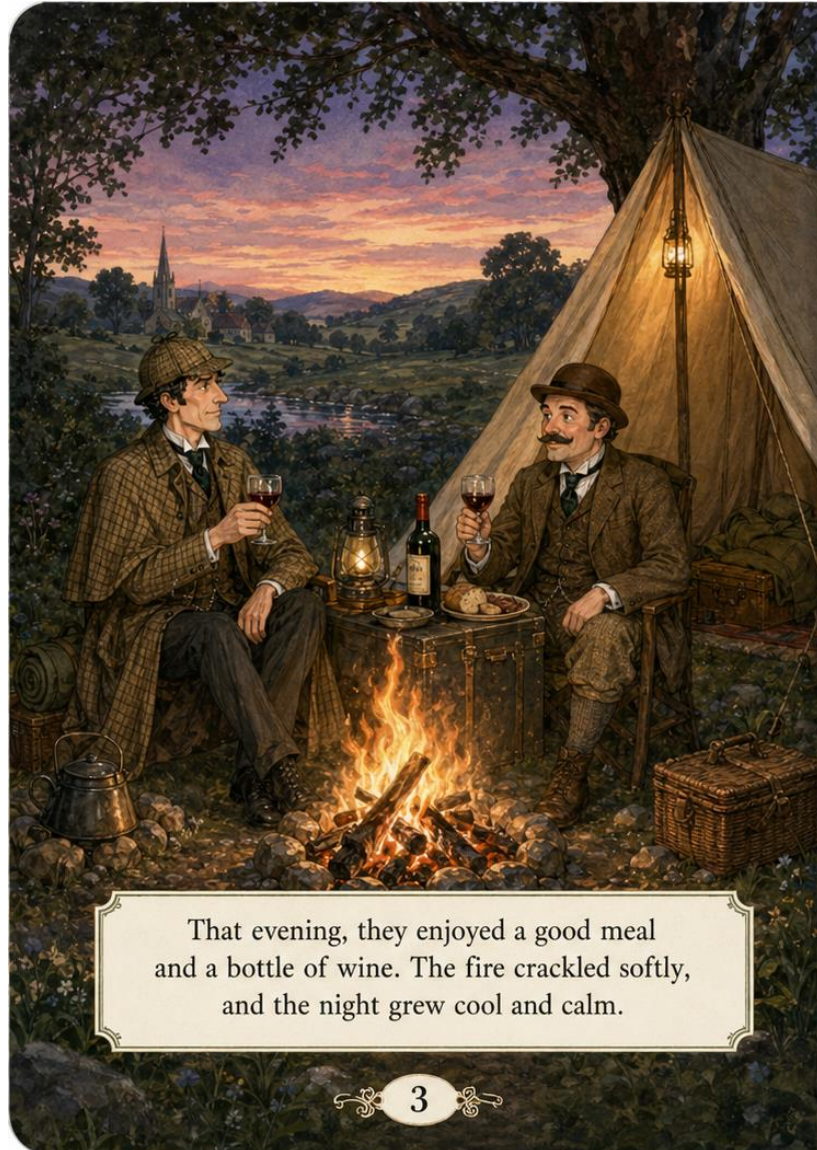
Comprehensive Hallucination Mitigation



Is MRAG Evaluation Sufficient?



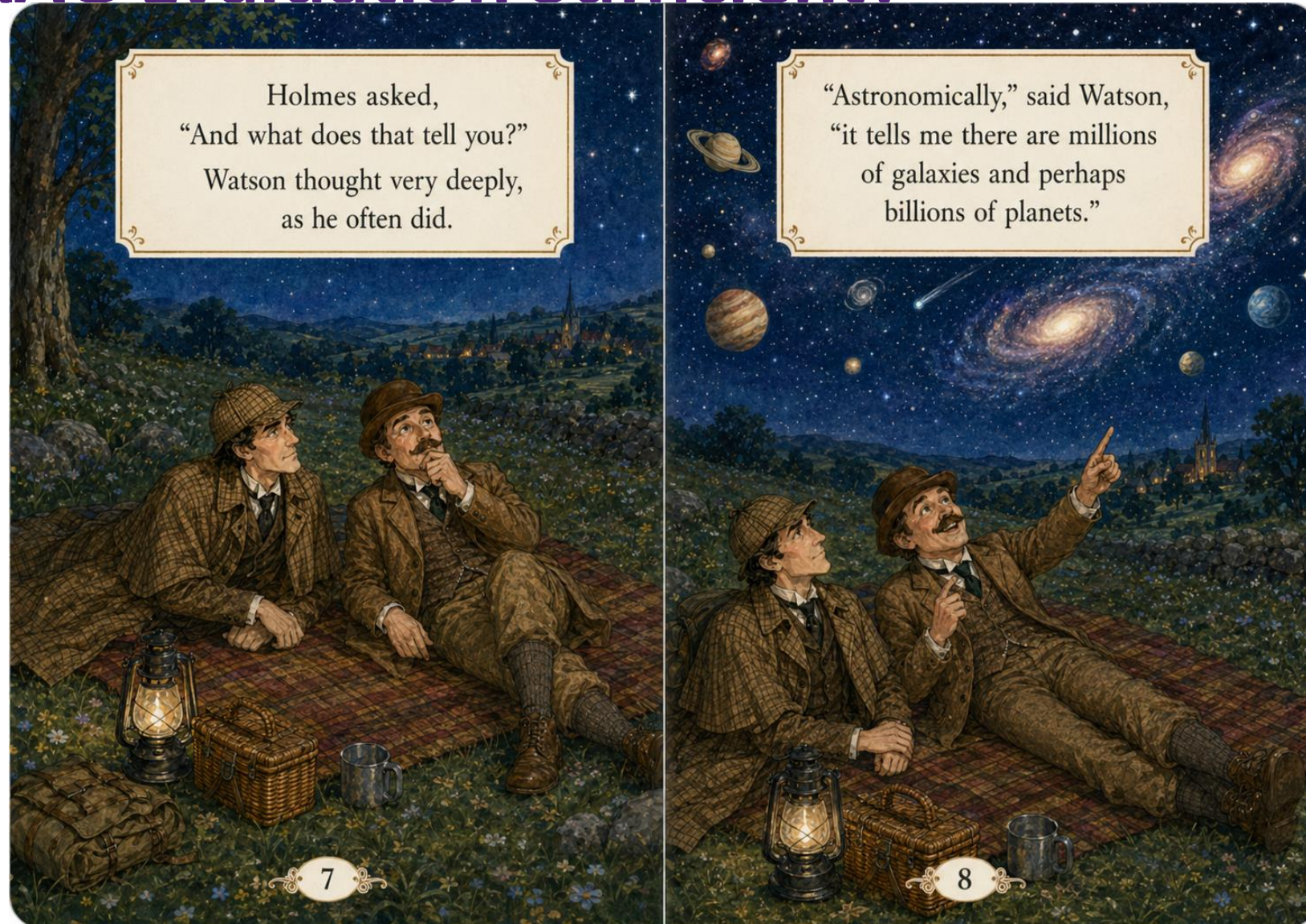
Is MRAG Evaluation Sufficient?



Is MRAG Evaluation Sufficient?



Is MRAG Evaluation Sufficient?



Is MRAG Evaluation Sufficient?



Is MRAG Evaluation Sufficient?



Is MRAG Evaluation Sufficient?



Is MRAG Evaluation Sufficient?

- The agent can observe many details correctly.
- The agent can even cite or describe evidence faithfully.
- But if it misses the missing tent, it has failed the task.
- This reveals a gap between **claim-level correctness** and **problem-level understanding**.

Did the system identify the evidence that actually matters?



What we covered today...

- **M-RAG needs layered evaluation.**

We reviewed retrieval quality, answer quality, and groundedness/faithfulness as three connected evaluation layers.

- **Retrieval metrics tell only part of the story.**

Precision, Recall, F1, MRR, and nDCG@K measure whether relevant evidence is retrieved and ranked well, but not whether it is used correctly.

- **Text-centric evaluation can fail in multimodal settings.**

A system may appear faithful from text alone while misreading the image, chart, radiology scan, layout, or query-side visual evidence.

- **Hallucination in M-RAG has multiple sources:** VCD, DPO/RL can help mitigate

Next Week: Agents

In multimodal RAG, the hardest failure may not be saying something false, but missing the obvious truth...



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THANKS!

Do you have any questions?

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